HIDDEN SYNERGY: HONG KONG'S ROLE IN THE DEVELOPMENT OF SCIENCE AND TECHNOLOGY IN CHINESE HIGHER EDUCATION

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ABSTRACT

This paper examines the rapid growth of China's science and technology (S&T) sector within the broader context of higher education, focusing on Hong Kong's crucial yet understudied role in this transformation. While conventional narratives emphasize China's relationship with the United States, our analysis reveals Hong Kong's distinct contribution as a vital bridge in China's scientific development. Drawing on comprehensive data from the Web of Science, institutional archives, university leader biographies, and PhD dissertation records, we identify three key patterns. First, Hong Kong ranks as one of mainland China's key international collaborators in S&T alongside established nations like the UK, Japan, and Germany, though the U.S. remains the dominant partner. Second, this collaboration shows clear temporal characteristics, with partnerships intensifying during the 1990s as Hong Kong's research doctoral programs matured and mainland students sought advanced training. Third, the collaboration exhibits distinct geographical patterns, with Beijing maintaining centrality despite distance, while Guangdong leverages its proximity to Hong Kong. Our findings challenge the perception of Hong Kong as merely a finance-driven city, revealing its role in fostering a robust academic community that facilitated knowledge transfer between mainland institutions and global scientific networks. This relationship thrived despite cultural and ideological differences, combining mainland China's motivated talent pool with Hong Kong's academic freedom and international connections. By documenting this understudied dimension of China's educational development, this paper offers new insights into the forces behind the country's emergence as a global scientific power.

Keywords higher education development • science and technology • Hong Kong • Mainland China • history of science

1 Introduction

In 2019, *The Economist* published "Can China become a scientific superpower?" The analysis acknowledged China's scientific achievements while expressing reservations about its global scientific leadership aspirations. The publication identified structural impediments: restricted academic freedom, questions regarding talent recruitment effectiveness, limited international recognition, emphasis on research quantity over quality, and constraints on international collaboration. These factors, the analysis concluded, would restrict China to pursuing rather than leading global scientific advancement (The Economist 2019).

By June 2024, *The Economist* reversed its assessment, declaring "China has become a scientific superpower." This revision reflected measurable progress: Chinese researchers had achieved prominence across scientific excellence indicators, demonstrating sustained progress in producing influential research. The established scientific hierarchy, historically dominated by American, European, and Japanese institutions, underwent significant reconfiguration (The Economist 2024). This transformation's exceptional speed warrants systematic examination.

What mechanisms enabled China's scientific advancement? Standard explanations emphasize the transformation of China's higher education system, particularly research universities. These institutions have expanded substantially since 1999, generating both skilled professionals and significant research contributions (Simon and Cao 2009; Hayhoe and Zha 2005). Observers note the scale and technological sophistication of Chinese universities (Rhoads et al. 2014). Scholars also frequently attribute this progress to Sino-American scientific collaboration, widely recognized as the most significant bilateral relationship in global science (Crow 2022; Hao and Hua 2023). This partnership has provided essential funding, training, and international exchange opportunities.

However, examining China's scientific development reveals a critical yet underexplored factor: Hong Kong's contribution. While existing scholarship emphasizes mainland industrial policies (Naughton 2024), university development (Freeman and Huang 2015), or U.S.-China cooperation (Crow 2022), Hong Kong has served as a fundamental yet understudied influence on China's scientific trajectory.

This role might appear unexpected, given Hong Kong's established reputation in finance and commerce rather than scientific innovation. Yet historical analysis demonstrates that Hong Kong functioned as an essential intermediary in China's scientific development. The city's distinctive geographic and institutional position enabled it to accelerate China's emergence as a global scientific leader during a crucial modernization period.

The foundations of this scientific partnership emerged in the 1970s as China began moving beyond isolation toward reform (Westad and Chen 2024). While mainland universities faced resource constraints and ideological pressures that limited advanced scientific training, Hong Kong's institutions flourished through Western academic connections, institutional independence, and robust funding. Scientific exchanges began before China's 1978 opening, as scientists, engineers, and medical professionals initiated mainland connections (Chan 2023; Chen 2016). Hong Kong's international port status facilitated visits by overseas Chinese scientists to the mainland. Following China's reforms, Hong Kong's universities became essential partners in developing mainland scientific capacity, providing advanced education, enabling intellectual exchange, and connecting with global scientific networks.

This collaboration reflected both practical considerations and shared objectives. Hong Kong's universities in the 1980s and 1990s sought to enhance their international standing while contributing to China's modernization—a goal many Hong Kong academics supported despite political differences with the mainland government. At the same time, university leaders saw fundamental limitation in the city own fate. The business tycoons did not recognize the importance of research and developments, while the small population inhibits the development of the S&T sector (Woo 2007). Across the Shenzhen river, Mainland leadership prioritized modernization, recognizing scientific advancement as crucial to China's global emergence (Westad and Chen 2024). This created productive exchanges: mainland China provided motivated, prepared students, while Hong Kong offered resources, training, and academic freedom.

Drawing on historical documentation, institutional records, and bibliographic data, this paper examines Hong Kong's pivotal role. Evidence indicates that Hong Kong maintained significant influence as mainland institutions developed, serving as an intellectual and operational bridge that accelerated China's contemporary scientific development. Its universities provided crucial support for Chinese scientists navigating mainland institutional constraints, offered essential resources during China's early scientific investment period, and established networks facilitating international collaboration.

By tracing the evolution of Hong Kong–mainland academic relationships, this analysis reframes China's scientific advancement narrative. Rather than attributing this progress solely to recent reforms or U.S.-China exchanges, we identify a deeper historical process. Hong Kong's role proves fundamentally transformative, expediting China's scientific development and establishing foundations for global leadership in science and technology.

2 China's Ascendance in Science and Technology

China's rise in science and technology (S&T) over the past four decades is nothing short of astonishing. Measured by scientific paper production, the transformation is dramatic. In the early 1980s, China's annual scientific output accounted for less than 1% of the world's total, using data from the Web of Science (WoS)—a reflection of the long-term damage inflicted by the Cultural Revolution, which disrupted education and research for over a decade. Yet, beginning in the 1990s, China's scientific production entered a phase of rapid expansion, tripling every decade. By 2023, China had risen to produce 37% of the world's scientific papers, surpassing every other nation and cementing its prominence in global S&T output.

But it is not just the scale of production that astonishes; China has also demonstrated remarkable progress in the quality of its research. Two key metrics illustrate this point. The first is the share of globally highly-cited papers, a measure of research influence. According to WoS, which publishes the data since 2014, China's contribution to the top 1% of highly-cited papers has grown from less than 20% levels in 2014 to an impressive 51% in 2023—rivalling and, in

some fields, surpassing the traditional scientific powerhouses of the United States (48.39% in 2014 to 29.41% in 2023) (Tollefson 2018; Wagner, Zhang, and Leydesdorff 2022).

Another perspective comes from the prestigious *Nature* magazine, which introduced the *Nature Index* in 2016 to evaluate research output in a select group of elite scientific journals. These journals are widely recognized as representing the pinnacle of scientific excellence. In 2016, China was already ranked second, trailing only the United States. By 2023, China overtook the U.S. to claim the top spot, a milestone that signifies its emergence as a global leader in cutting-edge research (Woolston 2023).

Supplementary data from other sources reinforces this narrative. For example, Japan's National Institute of S&T Policy's 2024 report placed China as either first or second globally in key metrics such as R&D expenditure, the number of researchers, and the number of scientific publications (Center for S&T Foresight and Indicators 2024). Similarly, the Hamilton Index, developed by the U.S.-based Information Technology and Innovation Foundation (ITIF), evaluates national competitiveness in advanced industries. In its 2023 report, the index described China as "dominating strategically important industries." (Atkinson and Tufts 2023, 1) The 2024 mid-year report stated in its title that "China is rapidly becoming a leading innovator in advanced industries," (Atkinson 2024) signaling the nation's growing influence in fields critical to the global economy.

The World Intellectual Property Organization (WIPO), which serves as a clearinghouse for global patents, has also recognized China's progress. In its 2022 report (Dutta et al. 2022), WIPO placed China "near the top 10" in patent innovation. This improvement is underscored by the sharp rise in Chinese patent filings and international applications, reflecting the country's commitment to protecting and commercializing its intellectual property.

Even skeptics and critics have acknowledged China's achievements. In 2024, then U.S. Senator Marco Rubio, a long-time China hawk, released a report concluding that China had made remarkable advances in nine strategic industries, including artificial intelligence, quantum computing, biotechnology, and advanced manufacturing (The Office of Senator Marco Rubio 2024). This acknowledgment highlights the strategic significance of China's scientific and technological ascendance on the global stage.

Taken together, these indicators depict a country that has not only closed the gap with traditional S&T leaders but has, in many cases, surpassed them. This remarkable transformation underscores China's strategic prioritization of S&T as a cornerstone of its modernization efforts, a vision that has been executed with remarkable speed and scale.

3 Data and Method

This paper presents an analysis of China's ascendancy through the examination of quantitative data and historical materials. Our research aims to advance the current understanding of this transformative process.

The study draws upon bibliographic data from Web of Science (WoS) to examine scientific collaboration patterns between Hong Kong, Mainland China, and the international academic community. To provide deeper context, we analyze archival materials from three prestigious Hong Kong institutions: The University of Hong Kong (HKU), The Chinese University of Hong Kong (CUHK), and The Hong Kong University of Science and Technology (HKUST). These institutional records include administrative reports and comprehensive graduate student information.

Our dataset encompasses 43 million entries from WoS, with 310,000 originating from Hong Kong (H) and 6 million from Mainland China (B). The historical materials include 13 personal biographies and 2 collections of memoirs, supplemented by university reports spanning from the 1980s through the 2020s. To deepen our analysis of academic development, we gathered S&T PhD dissertation data from institutional library websites: HKU contributed 6,691 dissertations (1990-2020), CUHK provided 4,226 dissertations (1990-2020), and HKUST added 4,463 dissertations (1995-2020).

These dissertation records, which include titles, authors' names, departments, and completion years, offer valuable insights into Hong Kong's academic development before and after strengthening ties with Mainland China. This comprehensive collection of records provides a detailed view of the evolution of higher education and scholarly output in Hong Kong during this pivotal period. Complete discussions appear in the appendix, offering readers access to the full scope of our source materials.

4 The Elephant and the Cub

How did China achieve its rapid rise in science and technology? One clear explanation lies in Sino-American collaboration. Since the 1979 signing of the U.S.-China Science and Technology Agreement, the two nations have engaged in increasingly robust cooperation across multiple scientific fields (Lampton, Madancy, and Williams 1986). This trend, which lasted well into the 2010s, has often been described as the most significant bilateral relationship in global

science and technology (Hao and Hua 2023). The U.S. played a vital role in helping China build modern research capabilities and China actively engaged in learning from the U.S. The data for this section is in Table A1.

4.1 The Scale of Sino-American Collaboration

As the latecomer and main beneficiary, China demonstrated remarkable growth in internationally co-authored scientific papers over the decades.[1] In 1980, only 9% of Chinese papers included any international co-authors (69 out of 767 from WoS, or b_2/B in our terminology). By 1990, this figure had doubled to 20.69% (1,489 out of 7,195), signaling China's deeper integration into global science networks. By 2000, the share stabilized at 23% (5,704 out of 24,776), maintaining similar levels through the 2010s. By 2020, just over one-quarter of all Chinese S&T papers had at least one international collaborator (135,659 out of 532,929).

The United States played a particularly prominent role. Using a narrow definition — papers with *only* U.S.-based co-authors (b_{2a}) — the data reveal a striking trend.[2] In 1980, 4.17% of Chinese publications included U.S. collaborators (b_{2a}/B). This share grew steadily to 7.51% in 1990 and fluctuated between 6–8% through the 2000s and 2010s. Collaboration peaked in 2017 when U.S.-based researchers contributed to 9.27% of all Chinese publications. By 2023, however, this share had dropped to 3.62%, reflecting a broader decline in Sino-American relations. While the percentage of collaborative publications peaked in 2017 and subsequently declined, the absolute number of collaborated publications grew dramatically—from just 32 papers in 1980 to 38,455 in 2020.

A broader measure—counting all Chinese papers with at least one additional international collaborator to the U.S. ($b_{2a} + b_{2b}$), that is, international collaboration *involves* U.S.—provides additional context. By this metric, U.S.-China collaborations accounted for 5.22% of Chinese publications in 1980 ($(b_{2a} + b_{2b})/B$), rising to 9.27% in 1990. The share was 7.64% in 2000, 10.30% in 2010, and peaked at 12.13% in 2017 before falling to 5.45% in 2023.[3]

4.2 The U.S. Share in China's International Collaborations

To understand Sino-American S&T collaboration within the broader scope of China's global partnerships, we examined the U.S. share of all Chinese papers with international collaborators. Using the narrower definition, U.S.-China collaborations represented 46.38% of China's internationally co-authored papers in 1980 (b_{2a}/b_2). This proportion steadily declined: 36.27% in 1990, 24.93% in 2000, rose again to 34.63% in 2010, and peaked at 37.26% in 2015, before falling sharply to 19.55% in 2023.

The broader definition reveals even higher numbers. By this measure ($(b_{2a} + b_{2b})/b_2$), U.S. collaborators accounted for 57.97% of such papers in 1980, 44.80% in 1990, 33.17% in 2000, 43.98% in 2010, and peaked at 48.19% in 2015, declining to 29.40% in 2023.

These figures highlight the unparalleled scale of Sino-American S&T collaboration, which has consistently outpaced other bilateral scientific partnerships. It remains the "elephant" in China's international scientific network.

4.3 The Rise of a Surprising "Cub": Hong Kong

While the Sino-American collaboration dominates the narrative, other actors have played key roles in assisting China's rise in S&T. Conventional suspects include industrialized nations like Germany, the United Kingdom, Japan, and Australia. However, an unexpected but critical contributor is Hong Kong, a "cub" that has emerged as China's significant partner in international collaboration.[4]

From a scale perspective, Hong Kong ranks as one of mainland China's key international collaborators in S&T alongside major Western nations. To be sure, research internationalization has always been central to Hong Kong's academic development in science and technology. In 1980, before establishing substantial doctoral programs or mainland

[1] In analyzes of research collaboration metrics, "international" refers specifically to research partnerships involving investigators outside mainland China. Research partnerships between mainland Chinese institutions and those in Hong Kong, Macao, and Taiwan are classified as international collaborations, consistent with Web of Science bibliometric standards.

[2] See Appendix A for detailed methodological discussion.

[3] While the percentage of collaboration in 2023 approximates that of 1980, the context differs substantially. In 1980, Chinese and American researchers jointly produced only 40 papers. By 2023, this number had grown to 37,930, representing a remarkable increase in absolute terms. However, China's overall S&T research output grew even more dramatically: from 767 papers in 1980 to 696,291 papers in 2023, demonstrating that China's total research production has outpaced the growth in Sino-American collaborative work.

[4] Hong Kong's role in China's S&T development is surprising given its small population and unique developmental trajectory. With a relatively small population, Hong Kong might seem an unlikely candidate for becoming a significant player in science and technology. Until the 1990s, Hong Kong lacked robust PhD programs and government infrastructure to support research and development. Accounts from students and faculty, such as Liu (1993) and Chan (2023), as well as perspectives from university leaders like Huang (2000) and Woo (2007), further illustrate these limitations.

Chinese partnerships, international collaboration[5] comprised 18.81% of Hong Kong's S&T output (h_{2o}/H). This proportion increased steadily over two decades, reaching 22.84% in 2000, then stabilized between 22-24% for the next twenty years, measuring 19.50% in 2023.

The entry of mainland China transformed Hong Kong's collaborative research landscape. During the 1980s, collaboration remained limited, with annual co-publications ($(h_{2a} + h_{2b})/H$) staying below 10%, reflecting both Hong Kong's developing research infrastructure and mainland China's recovery from earlier disruptions. By 1990, mainland partners appeared on less than 4% of Hong Kong research papers. This proportion then grew steadily: 27.68% in 2000, 39.99% in 2010, reaching 60.70% in 2020. As of 2023, this upward trend continues unabated.

Hong Kong's research collaboration extends beyond bilateral partnerships with mainland China. A notable pattern emerges in three-way collaborations, involving authors from Hong Kong, mainland China, and at least one other international location (h_{2b}). This pattern developed gradually: in 1997, prior to the handover, such papers numbered only 83 (2.29% of Hong Kong's total output). The volume increased substantially thereafter: 254 papers in 2000, expanding to 4,699 in 2020, and reaching 6,876 papers in 2023 (28.43% of total output). Since 2018, these three-way collaborations have exceeded traditional international collaborations (papers with Hong Kong and international authors but no mainland Chinese authors, h_{2o}). This trend suggests Hong Kong's role as a collaboration center bridging mainland Chinese scholars with the international academic community.[6]

5 Examining the Hong Kong–Mainland China Collaborative Patterns in S&T

Looking beyond general trends to individual institutions reveals new and intriguing patterns that reflect how S&T research operates in practice.

5.1 Key Collaboration Locations for Mainland Universities

To analyze the collaborative relationships between Hong Kong and mainland institutions, we picked 62 Chinese universities from the 2024 edition of the *Nature Index*.[7] For each university, we pooled their publication records from 1980 to 2023 and calculate the percentage of its international collaborators. This analysis highlights the dynamic nature of partnerships between Hong Kong and various regions in China.

Figure 1 presents a dual-panel analysis of international research collaboration patterns. The upper panel examines 45 universities identified as leading S&T institutions by the *Nature Index*, while the lower panel focuses on 17 universities within Guangdong province. The analysis tracks collaboration patterns across 11 major international research partners.[8]

If we only look at the Top 45 universities (i.e. the upper panel), the data reveals the United States as the dominant international collaborator, ranking first for 44 of the 45 institutions analyzed. However, Hong Kong's position among established S&T nations like the UK, Japan, and Germany warrants particular attention. Shenzhen University stands out by ranking Hong Kong as its primary collaborator, while 16 universities position Hong Kong as their second most frequent collaborator, and 7 place it third. This strong showing is remarkable considering Hong Kong's smaller scale and relatively recent research development.

The upper panel, which orders universities by their collaboration intensity with Hong Kong institutions, reveals a clear geographic pattern. The top five positions are occupied by Guangdong universities. Guangdong's leading institutions—Sun Yat-sen University, Southern University of Science and Technology, Shenzhen University, and Jinan University—demonstrate particularly strong collaborative ties with Hong Kong, suggesting they effectively utilize their geographic

[5] Hong Kong's research output comprises four categories: (1) papers with only Hong Kong authors (h_1), (2) papers with exclusively mainland China collaborators (h_{2a}), (3) papers with both mainland China-based authors and authors from other countries/regions (h_{2b}), and (4) papers with international collaborators excluding mainland China (h_{2o}). Here we refer to the last category, h_{2o}.

[6] While this descriptive indicator offers valuable insights, it cannot support causal claims, allowing only for careful conjectures. Our evidence, presenting in the following sections, suggests a temporal pattern: Hong Kong developed its S&T research capabilities approximately a decade before mainland China. This chronology makes it difficult to support arguments that position mainland China in a bridging role, as such interpretations would require additional assumptions. Following Occam's razor, the simpler explanation of direct bilateral development appears more plausible.

[7] See Table B1 for the complete list in the Appendix.

[8] This analysis builds on the idea of "proportional collaboration" as there is no perfectly mutually exclusive way to categorize collaboration when authors from multiple countries contribute to the same paper. To quantify proportional collaboration we calculate a collaboration score, see Appendix A.

In addition to Hong Kong, international research partners include: Australia, Canada, England, France, Germany, Japan, Russia, Singapore, South Korean, and the U.S.

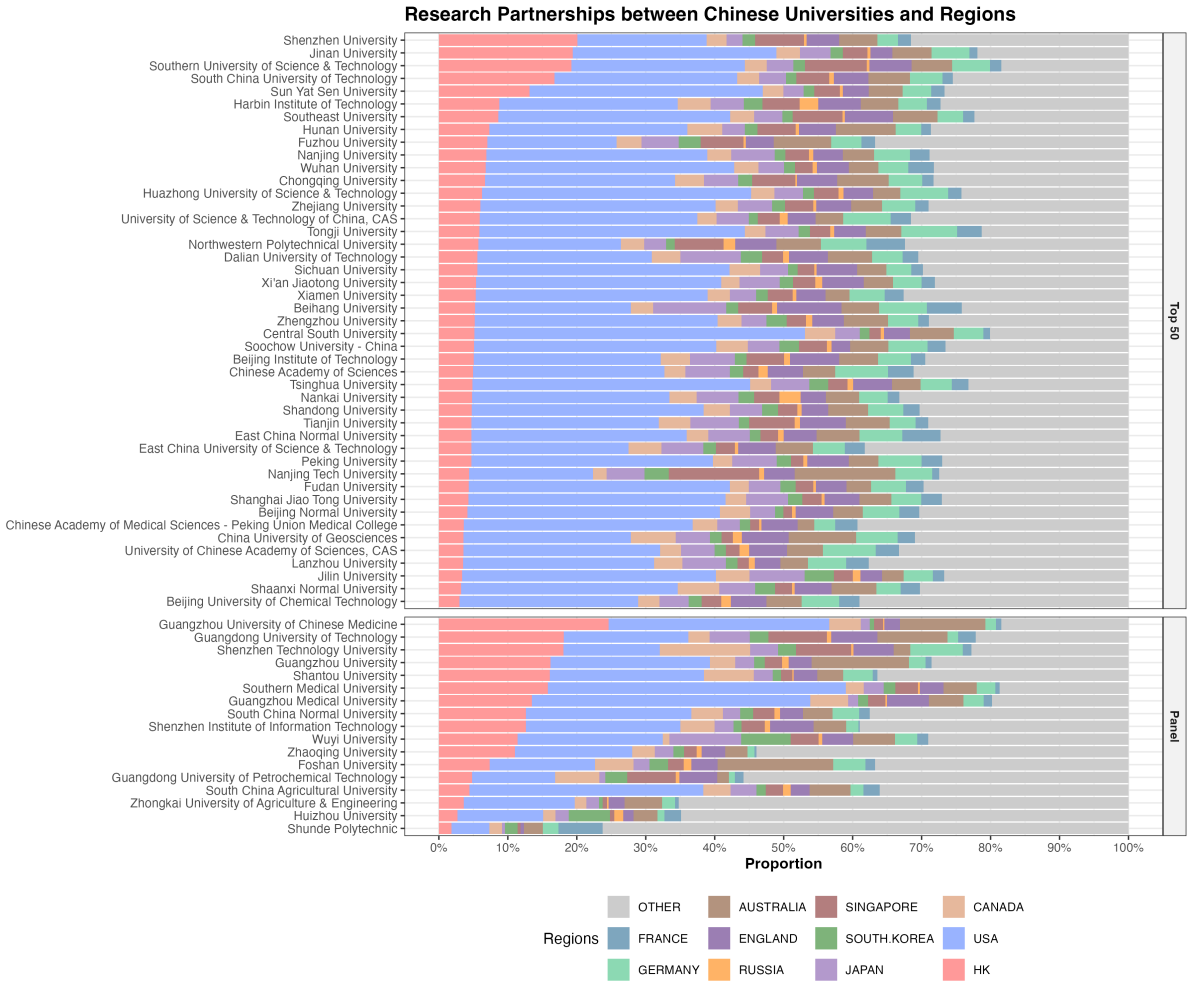


Figure 1: Mainland Chinese universities' international collaboration

proximity. Hong Kong maintains significant collaborative relationships with mainland China's premier institutions, albeit at lower frequencies than these regional partnerships (i.e. the U.S.'s role is more pronounced).

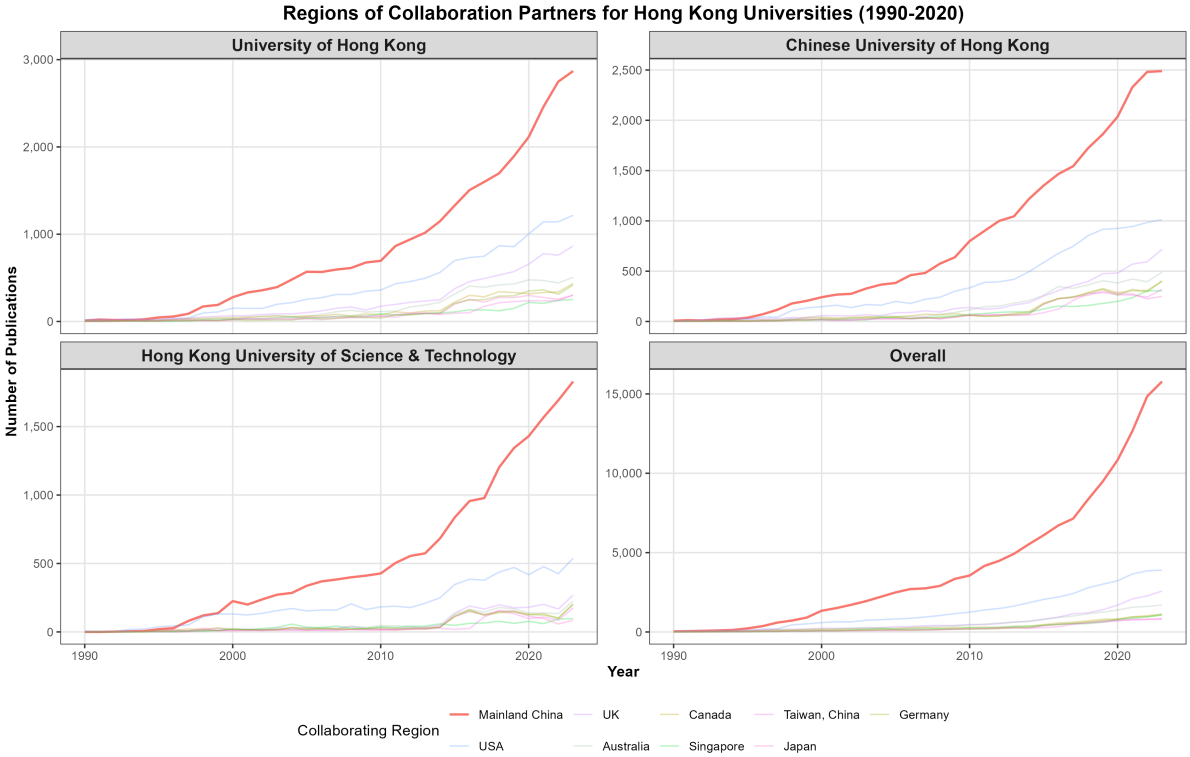
Regional patterns reveal stronger collaborative ties between Hong Kong and universities in Guangdong province. Among the 22 Guangdong institutions identified in Table B1, only five rank among China's top 45 S&T universities, reflecting Guangdong's historical limitations in higher education despite its economic strength. Within this group, two universities rank Hong Kong above the U.S. as their primary collaborator, 17 position Hong Kong as their second most frequent partner, and three list Hong Kong third.

5.2 Key Collaboration Locations for Hong Kong Universities

Building on the previous analysis, a key question arises: is the partnership between Hong Kong and mainland institutions symmetrical? Specifically, do mainland institutions mainly play a supporting role in their scientific collaborations with Hong Kong, or do they hold a more central, reciprocal position, as suggested by $h_{2a} + h_{2b}$ number?

The answer is no. The evidence demonstrates a clear pattern. At the university level, collaboration with mainland institutions is not only reciprocal but primary. Mainland universities have become major contributors to Hong Kong's research output, establishing themselves as principal international collaborators. As illustrated in Figure 2, which shows the top international collaborators for three Hong Kong institutions, mainland institutions have surpassed traditionally dominant regions to become Hong Kong's primary research partners. This shift took place in the mid to late-1990s (CUHK: 1995, HKU: 1994, HKUST: 1997) and has strengthened consistently, as next section will show, demonstrating the durability of this cross-border academic relationship.

For Hong Kong's universities, this orientation toward mainland partnerships indicates a significant transformation, where mainland collaborations have become essential to Hong Kong's standing in global academia. While Hong Kong universities maintain worldwide collaborations, their S&T research output depends substantially on connections with mainland institutions. Mainland universities now occupy positions previously held by partners like the United States and Europe, offering sustained research opportunities enhanced by shared cultural and linguistic foundations.



Notes: Collaborating Regions denote the origins of those co-authors collaborated with the targeted institution

Figure 2: Regions of collaboration partners for Universities in Hong Kong (1990-2015)

From the mainland perspective, as seen in Figure 1, Hong Kong serves as a valuable but supplementary partner within a broader network of international collaborations, particularly with the U.S. While top mainland institutions value Hong Kong for advancing scientific research and expanding S&T initiatives, the relationship holds different significance than it does for Hong Kong universities. This creates a distinct pattern: mainland universities maintain diverse international partnerships, while Hong Kong universities show increasing reliance on mainland institutions as primary research partners and bridges. This arrangement strengthens Hong Kong's academic position while providing mainland universities with a strong regional collaborator.

This pattern reflects broader regional developments, where Hong Kong's universities employ their autonomy, resources, international reputation, and geographic proximity to build productive partnerships with mainland institutions. Beyond formal agreements or joint projects, the academic relationship has become central to Hong Kong universities' research priorities and strategic planning. While mainland institutions maintain global engagement, Hong Kong universities find in mainland China an essential research partner, making this relationship fundamental to their position in global academia.

5.3 Major Collaborative Subject Areas between Hong Kong and Mainland China

Universities in Hong Kong demonstrate significant strategic autonomy in selecting and developing their research collaborations with mainland institutions. This section examines the key research areas where Hong Kong and mainland China collaborated between 1990 and 2023, a period that saw Hong Kong emerge as a crucial international partner in China's scientific development.

Scientific research classification presents inherent complexities due to overlapping disciplines and evolving fields. To address this, we developed a systematic classification approach. Our analysis of publications from three Hong Kong

universities identified 112 research topics in Web of Science (WoS), including 110 classified topics and 2 unclassified categories (Table B2).

Our methodology consisted of two key steps: identifying each publication's primary research topic and developing a meso-level classification system that consolidated these topics into 27 broader categories (Table B3). For each category, we calculated the proportion of papers with mainland Chinese collaborators, expressed as $(h_{2a} + h_{2b})/H$. This analysis spanned three universities across three time periods: 1991-2000, 2001-2010, and 2011-2023 in Figure 3. There are four findings.

Overall Trends Across Fields: The data reveals a comprehensive shift toward increased mainland collaboration across virtually all S&T research domains, with the most rapid increases occurring in the 2011-2023 period. This trend is particularly evident in biomedical fields, where collaboration rates frequently exceeded 50% in the most recent period. The rise in collaboration isn't uniform but follows a pattern of gradual increase from 1991-2000 to 2001-2010, followed by a sharp acceleration in 2011-2023. Notably, even traditionally less collaborative fields like Agriculture & Food Systems and Infection & Immunity show substantial increases, suggesting a broadening of the collaboration base beyond just a few specialized areas. This pattern indicates a systematic rather than opportunistic approach to cross-border collaboration, likely reflecting both institutional policies and broader geopolitical developments.

Field-Specific Variation in Collaborative Patterns: Clinical Medicine & Support Sciences, Information Systems, and Engineering Systems emerge as particularly notable areas, showing both high volume of collaboration and consistent growth across all three universities. These fields exhibit some of the highest collaboration rates, often exceeding 60% in the 2011-2023 period. However, some fields show unexpected patterns: Geosciences displays a dramatic surge in collaboration rates at CUHK and HKU but more modest growth at HKUST, while Energy Technologies shows varying patterns across institutions, with HKUST maintaining relatively stable collaboration rates even as other universities increased theirs. Alternative Medicine presents another interesting case, showing particularly strong growth at CUHK and HKU but limited presence at HKUST, which does not have a medical school, suggesting institutional specialization in collaboration strategies.

Differences Between Universities: Each university exhibits distinct collaborative patterns that likely reflect their institutional strengths and strategic priorities. CUHK shows particularly strong collaboration growth in agriculture & food systems and alternative medicine, areas where it historically had strong programs. HKU demonstrates more balanced growth across fields but shows exceptional increases in clinical and medical sciences, aligning with its historical strength in medicine. HKUST, as a younger institution with a traditional focus on science and technology, shows the most dramatic increases in engineering systems and information systems, while also developing strong collaborations in environmental health and analytical instrumentation. These patterns suggest that while all three universities increased their mainland collaboration, they did so in ways that amplified their existing institutional strengths rather than converging on a single model.

Temporal Changes and Key Periods: The temporal evolution of collaboration shows three distinct phases. The 1991-2000 period generally shows modest collaboration levels (typically below 25%) across most fields, reflecting the early stages of cross-border academic integration. The 2001-2010 period shows a gradual increase in collaboration rates, typically reaching 25-35% in many fields, suggesting a period of systematic development of collaborative relationships. The 2011-2023 period marks a dramatic acceleration, with collaboration rates frequently doubling from the previous decade. This acceleration is particularly notable in specific fields: information systems, clinical sciences, and engineering systems show some of the most dramatic increases during this period. The timing of these changes suggests that while the groundwork for collaboration was laid in the earlier periods, the most recent decade saw the full flowering of these relationships, possibly influenced by both institutional policies and broader geopolitical developments.

6 Explaining the role of Hong Kong

Up to this point, we have demonstrated that research collaboration between Hong Kong and mainland China is both reciprocal and significant. This collaboration varies across geographic regions and shifts over time, revealing an evolving pattern. But a key question remains: why does this pattern exist? To address this, we offer three explanations: (1) the role of key individuals who served as crucial points of contact, (2) the timing of developmental changes in Hong Kong's higher education sector, and (3) the strategic location of Hong Kong as a bridge for mainland China's scientific development.

6.1 Points of Contact

The collaboration between Hong Kong and mainland China in S&T has been significantly shaped by key individuals who facilitated connections. On the mainland side, Chinese scientists began welcoming international visitors as early as the mid-1970s, even before the end of the Cultural Revolution (Suttmeier 1980). The Chinese scientific diaspora played a crucial role in this development. Following the Nixon Shock in 1971, prominent overseas Chinese scientists,

Figure 3: Top research areas collaborating with mainland China for three universities in Hong Kong

such as Nobel laureate Yang Chen-Ning, visited China, discovering both a damaged scientific infrastructure and a generation of Chinese S&T talents eager to learn (J. Yang 2016; Lubkin 1971). It is important to note that, while much attention has been focused on Yang—who was the first to visit the PRC in 1971—he was not alone. In fact, the list of such scientists is extensive. For example, a 1974 compilation of scientific visits by this group (Ho, Yang, et al. 1974) includes Wang Hao (mathematician/computer scientist), Zhu Zhaoxiang (physicist, semiconductor engineering), Niu Man-Chiang (biologist), Wu Chien-Shiung (physicist), Yuan Chia-Liu (physicist), and Chang Chieh-Chien (meteorologist).

These renowned scientists, many of whom had established themselves abroad, became conduits for collaboration, creating channels to assist Chinese scientists and providing valuable *feedback loops* between mainland institutions and their overseas counterparts. The two Nobel laureate, Yang Chen-Ning and Lee, Tsung-Dao, both participated in science policy debates and institutional building since 1972 (C.-N. Yang and Weng 2018, 64ff; Lee 1999, 17ff), though on separate occasions. Through their efforts, knowledge and practices circulated in both directions, enhancing the capacity of Chinese institutions while keeping them connected to global scientific advancements.

From the 1970s to the late 1980s, Hong Kong did not develop large-scale S&T research. The two flagship institutions—HKU and CUHK—had very small S&T departments. Chen Fong Ching, who began teaching at CUHK's physics department in 1966 and later became the university's Council Secretary in 1980, commented that "there is no foundation for science in Hong Kong" (Chen 2016, 71) and later quoting then HKU president Rayson Huang that "the Hong Kong government did not encourage or finance research at all" (Chen 2016, 75). In 1967, the CUHK physics department had 15 faculty members, 14 of whom were lecturers, with only one professor (Chen 2016, 73). This view was not unique. Zhang Youqi, who joined HKU's engineering department in 1977 and later became HKU's vice president in 1988, also told a biographer that the department began focusing on research only in the 1980s (Chan 2023, 80).

No surprise, then, that shortly after Deng Xiaoping announced the "spring of science" in 1978, mainland China first turned to the U.S. for learning. Chinese Nobel laureate Lee Tsung-Dao's China-U.S. Physics Examination and Application (CUSPEA) program, established after his visit to China, played a key role in this shift. In collaboration with top mainland institutions like CAS, PKU, and Tsinghua, CUSPEA sent over 900 Chinese physics students to U.S. universities between 1979 and 1989 (Shen and Zhao 2001), helping to provide early international exposure for Chinese researchers. There are many such exchanges took place during the 1980s (Lampton, Madancy, and Williams 1986). In early 2000, the U.S. is the country that host most Chinese students (Freeman and Huang 2015).

In 1980s, formal mechanisms for such exchanges were lacking. For instance, the mainland's primary funding agency, the China Scholarship Council, would not be established until 1996.[9] As a result, points of contact relied heavily on personal connections and institutional ties (i.e., Shaw (2013, 145)). In addition to Lee's initiative, many overseas Chinese scientists were individually approached by mainland institutions. Figures like physicist Woo Chia-Wei (who later became the founding president of HKUST) and plant biologist Kung Shain-dow (later Dean of Science at HKUST) regularly hosted Chinese visitors (Woo 2007, 35; Kung 2000, 110). This relatively small group of prominent overseas Chinese scientists (Ch'i 2014, 18)—many of whom were politically inclined to support mainland China (Shaw 2013, 143)—became pivotal facilitators of exchange.

Among these figures, Woo Chia-Wei stands out as an important case. Woo's first visit to the PRC was in 1978, during a four-month sabbatical, when he visited Fudan University and the Chinese Academy of Sciences (CAS) (Woo 2007, 35ff). At the time, he was a faculty member at Northwestern University, and he arranged for Chinese physicists to visit his institution, fostering early ties. Although Woo later became President of San Francisco State University (SFSU), his eventual return to Hong Kong as the founding president of HKUST (1991–2001) rekindled his mainland connections. In his second year as president, Woo led HKUST's management team on a visit to the mainland (Ch'i 2014, 226), initiating a series of bilateral exchanges and collaborations that strengthened ties.

HKUST was not alone in reaching out to mainland China; the mainland extended its invitation to oversea scientists, those in Hong Kong included, even earlier. In 1979, after the first international conference on particle physics was held in Conghua, Guangdong province, a group of attendees—including Yang Chen-Ning, Lee Tsung-Dao, and most overseas Chinese physicists—was invited to Beijing to meet Deng Xiaoping, Hua Guofeng, Qian Xuesen, and Qian Weichang (Chen 2016, 77). Later, in June 1982, the presidents of HKU and CUHK, Prof. Rayson Huang (1972-1986) and Prof. Ma Lin (1978-1987), were both invited to Beijing to meet Deng Xiaoping (Dai 1997, 117; Huang 2000, 119). Huang later recounted this visit to HKU students:

> [Deng Xiaoping] also discussed three key characteristics of the University of Hong Kong that should be maintained in the long term. First, it offers English-language instruction, which facilitates international academic exchange. Second, the university has foreign professors, and he anticipated that

[9] This does not mean that state-supported study abroad did not exist. The *danwei*-based allocation system was in place. However, ordinary students—outside of a handful of institutions—were not able to apply for this financial support.

> China's future development would require many such talents, making the experience of the University of Hong Kong highly valuable for reference. Third, the University of Hong Kong serves as an excellent platform for mainland scholars to interact with their Hong Kong counterparts, allowing for a better understanding of the international academic community. (Chan 2023, 13)

Therefore in mid 1980, current and future university leaders from Hong Kong have all visited mainland. Though there is no direct evidence to show that HKU and CUHK and the governing Chinese elites in Hong Kong fully implemented Deng's vision, universities in Hong Kong did pivot toward conducting research and recruiting more prominent overseas scholars during the 1980s. This shift was partly driven by the city's development and the governing elites' growing recognition of the importance of S&T and the untapped potential of overseas talent (Woo 2007; Chien 2024). The most well-known cases include Yang Chen-Ning, who joined CUHK as Professor-at-Large in 1986, Fields Medalist mathematician Yau Shing-Tung in 1993, and Turing Awardee Yao Chi-Chih, who followed in 2005.[10] Together they attracted mainland students and further solidified academic ties.

The establishment of HKUST in 1991 marked a significant expansion of Hong Kong's research capacity. Planning for the university began in 1988, with the first students admitted in 1991 (Chien 2024).

The development of Hong Kong's scientific institutions paralleled the experiences of Chinese scientists who pursued advanced degrees abroad during the Cold War. These scholars shared a vision of modern China with science and technology as a unifying foundation. As relations between East and West improved and economic growth accelerated—first in the Asian Tiger economies and later in mainland China—these scientists committed themselves to national development.[11]

This perspective was articulated by Ma Lin, president of CUHK, in his 1984 address to graduating students:

> For China to become truly modernized, it would not be enough to simply increase investments and introduce new technology. She would have to achieve no less than the transformation of her ethos and of her institutional framework ... adopting a long-term open-door policy ... determined to learn from others and seek change in the most fundamental way.

The movement of Chinese scientists between countries demonstrates a complex pattern of professional and national relationships. While some overseas Chinese scientists who visited China chose to return permanently, others maintained international careers. These decisions frequently intersected with political circumstances, creating diverse professional trajectories.[12]

Within this context, Hong Kong emerged as a strategic location, offering scientists the opportunity to maintain connections with mainland China while preserving professional autonomy.[13] The cases of Yang, Yau and Yao, who eventually relocated to mainland China in the 2000s, exemplify one path. However, the pattern varies significantly, as other scientists either briefly worked in Hong Kong or remained there without pursuing mainland opportunities.

6.2 Temporal Characteristics

The establishment of connections between Hong Kong and mainland China by Chinese scientists therefore represents a significant historical process spanning multiple decades. This development also exhibits distinct temporal characteristics, as noted by Kung Shain-dow in his memoir about HKUST:

[10] Yang first visited the newly built university in 1964, giving a public lecture that caused a city-wide sensation (J. Yang 2016, 334; Chen 2003, 4). He visited more frequently starting in 1979 (Chen 2016, 199). Yau previously declined a position at HKUST in 1988, arguing that "Hong Kong will never be able to establish a research-oriented university. Brilliant professors won't return; those who do are mostly people who couldn't make it in the U.S." (Chien 2024, 18). Their distinguished professorship is documented by the CUHK library. See: https://www.lib.cuhk.edu.hk/en/libraries/ugallery/zonec/scholars/

[11] The parallel between scientific and cultural unification emerges clearly in Chen Fong-Ching's address to Yang Chen-Ning. Chen drew a profound connection between Yang's work on gauge field theory and broader social unity: "if the unifying power of the gauge field theory is remote to most of us, the need for greater unification among different worlds must be ever more keenly felt by all of us today, not only because of the strife and conflict that continue to divide and inflame the world like half a century ago, but also because of our own ceaseless traveling and migration for a thousand reasons in both geographic and cyberspace, which certainly far exceed what those New World-bound students of the bygone days could have imagined" (Chen 2003, 6). For an excellent account of economics explanation in East Asia, see Wade (2018).

[12] Political considerations influenced scientific work in both nations. See Wang and Hu (2014) and Wang (2022) for the Chinese context and Wang (2002) for the American perspective.

[13] Hong Kong's role as an intermediary extended beyond mainland-overseas connections. A significant milestone occurred in 1992 when the first mainland Chinese scientific delegation visited Taiwan since 1949. This exchange was orchestrated by Kung Shain-dow, who initiated discussions in the United States and continued his mediating role after relocating to Hong Kong (Kung 2004, 61).

> [The returnees] were shaped by their era—without the civil war over 50 years ago, this group would not have existed. Most of them were born in Mainland China, spent their childhoods in Taiwan during the civil war, and later pursued university education there before going to the United States for further studies. After completing their doctoral degrees in the U.S., most of them worked in teaching and research positions at universities or research institutions, achieving significant contributions in their respective fields. Most importantly, they remained deeply committed to China, and when the opportunity arose, they returned. Returning to Hong Kong to serve their country allowed them to fulfill their patriotic aspirations without the burdens often associated with it. Had HKUST been founded five years earlier, these individuals might not have been ready to return; had it been founded five years later, many would have been nearing retirement age. (Kung 2004, 5)[14]

Beyond the common motivations Kung described, competitive compensation played a crucial role in attracting faculty members. The Hong Kong government implemented substantial salary increases for university professors in two phases: first doubling salaries between 1988 and 1993, then raising them by an additional 170% from 1993 to 1998. This cumulative increase of 270% over the decade significantly enhanced Hong Kong universities' ability to recruit international talent (Kung 2004, 5).

Apart from the salary increase, research funding structures underwent significant changes in the 1990s. Initially, universities received block grants from the government, with total financial appropriations reaching 20 million Hong Kong dollars for all the institutions in Hong Kong. This amount was notably smaller than the research budgets of major U.S. universities (Woo 2007, 326).

While philanthropic donations provided additional support to universities (Huang 2000; Kao 2011; Woo 2007), the establishment of the Research Grant Council (RGC) in 1991 represented a fundamental shift in research funding. The RGC introduced a competitive research grant system that particularly benefited science and technology research, which required substantial financial resources compared to the social sciences and humanities research traditionally associated with Hong Kong (Chien 2024, 177; Ch'i 2014, 39ff).

Table 1 shows annual RGC funding by two broad categories: Science and Technology (includes Engineering, Biomedical Sciences, and Physics Sciences) and Social Sciences (includes humanities, Social Sciences, and Business studies). The rapid growth of research funding in S&T fields in the first seven years provides another piece of evidence that the city's research is accelerating during the 1990s.

This growth trajectory encountered two notable disruptions: the Asian Financial Crisis of 1997 and the SARS outbreak of 2003. Despite these challenges, the data shows resilience in Hong Kong's research funding commitments. The interim period between these events (2000-2003) maintained stable funding levels, while the post-2006 period marked a return to steady growth in S&T investment. Regardless of these disruptions, Hong Kong government consistently allocated 80% or more of its annual research fund to S&T fields during 1991 to 2010, reinforcing Hong Kong's commitment to science and technology research advancement.

The recruitment of accomplished faculty members and stable research founding contributed to the expansion of research doctoral programs. While comprehensive PhD admission data for individual universities remains unavailable, graduation statistics provide valuable insights. Based on the dissertation database in three universities, we calculate S&T graduates (i.e. students graduate from any departmental level unit in Faculty of Science, Faculty of Engineering, and Faculty of Medicine).

[14]Though not as elegant, similar remarks are made by Kung's colleague, historian Ch'i Hsi-Sheng (2014, 21ff).

Table 1: Annual Research Grant Allocation (in Thousands of HKD)

Year	S&T		Social Sciences+Humanities+Bus		Total	
	annual (ths)	year-to-year growth	annual (ths)	year-to-year growth	annual (ths)	year-to-year growth
1991	53,304	–	10,490	–	63,794	–
1992	67,495	27%	11,914	14%	79,409	24%
1993	86,927	29%	13,057	10%	99,984	26%
1994	155,601	79%	37,255	185%	192,856	93%
1995	172,149	11%	33,399	-10%	205,548	7%
1996	214,040	24%	40,300	21%	254,340	24%
1997	262,192	23%	60,548	50%	323,740	27%
1998	261,527	-1%	68,084	12%	329,611	2%
1999	262,216	0%	62,281	-9%	324,497	-2%
2000	294,468	12%	74,138	19%	368,606	14%
2001	337,911	15%	65,012	-12%	402,923	9%
2002	353,186	5%	74,800	15%	427,986	6%
2003	387,299	10%	75,472	1%	462,771	8%
2004	333,114	-14%	70,382	-7%	403,496	-13%
2005	327,280	-2%	77,295	10%	404,575	0%
2006	403,536	23%	85,790	11%	489,327	21%
2007	402,033	0%	84,962	-1%	486,995	0%
2008	412,024	2%	100,425	18%	512,449	5%
2009	417,783	1%	99,249	-1%	517,032	1%
2010	434,417	4%	110,163	11%	544,580	5%
2011	435,450	0%	122,029	11%	557,479	2%
2012	404,540	-7%	119,225	-2%	523,765	-6%
2013	430,664	6%	129,962	9%	560,626	7%
2014	449,334	4%	145,549	12%	594,883	6%
2015	448,104	0%	148,438	2%	596,542	0%
2016	455,692	2%	143,509	-3%	599,201	0%
2017	440,300	-3%	153,077	7%	593,377	-1%
2018	436,340	-1%	155,902	2%	592,242	0%
2019	482,206	11%	172,286	11%	654,492	11%
2020	575,265	19%	202,088	17%	777,353	19%
2021	580,785	1%	199,857	-1%	780,642	0%
2022	725,960	25%	255,504	28%	981,464	26%
2023	728,262	0%	250,885	-2%	979,147	0%
2024	740,866	2%	240,590	-4%	981,456	0%

With the help of detailed dissertation data, we can trace the growth of PhD graduates. We process each dissertation author name to estimate the author's origin. That gave us two variables: whether the students are mainland or not; whether the students study in S&T field or not.

As shown in Figure 4, Hong Kong universities produced few PhDs before the 1990s. HKU and CUHK each graduated fewer than 50 PhDs annually, indicating their primary focus on teaching rather than research during this period. In 1990, HKU graduated 38 PhDs and 24 of them were in S&T fields. Similarly, CUHK graduated 11 PhDs in the same year and 7 of them were in S&T fields. HKUST's first PhD graduate was in 1994. Not surprisingly, as the university's name suggests, the majority of its early PhDs were in S&T. [15]

[15] In retrospect, HKU, CUHK, and HKUST became "Science and Technology" universities because related departments graduated most students throughout their history. The PhD graduate numbers confirm this point. By our calculation, the three schools of Engineering, Science, and Medicine at HKU produced a combined 65% of all PhDs. CUHK graduated a similar percentage of PhDs in these disciplines: 68%. HKUST graduated an even larger percentage: 85% for just the Science and Engineering departments. Even if we exclude Medicine, the numbers are equally impressive: 46% for HKU and 44% for CUHK. These PhD graduate numbers show the historical emphasis on scientific and technological education at all three universities.

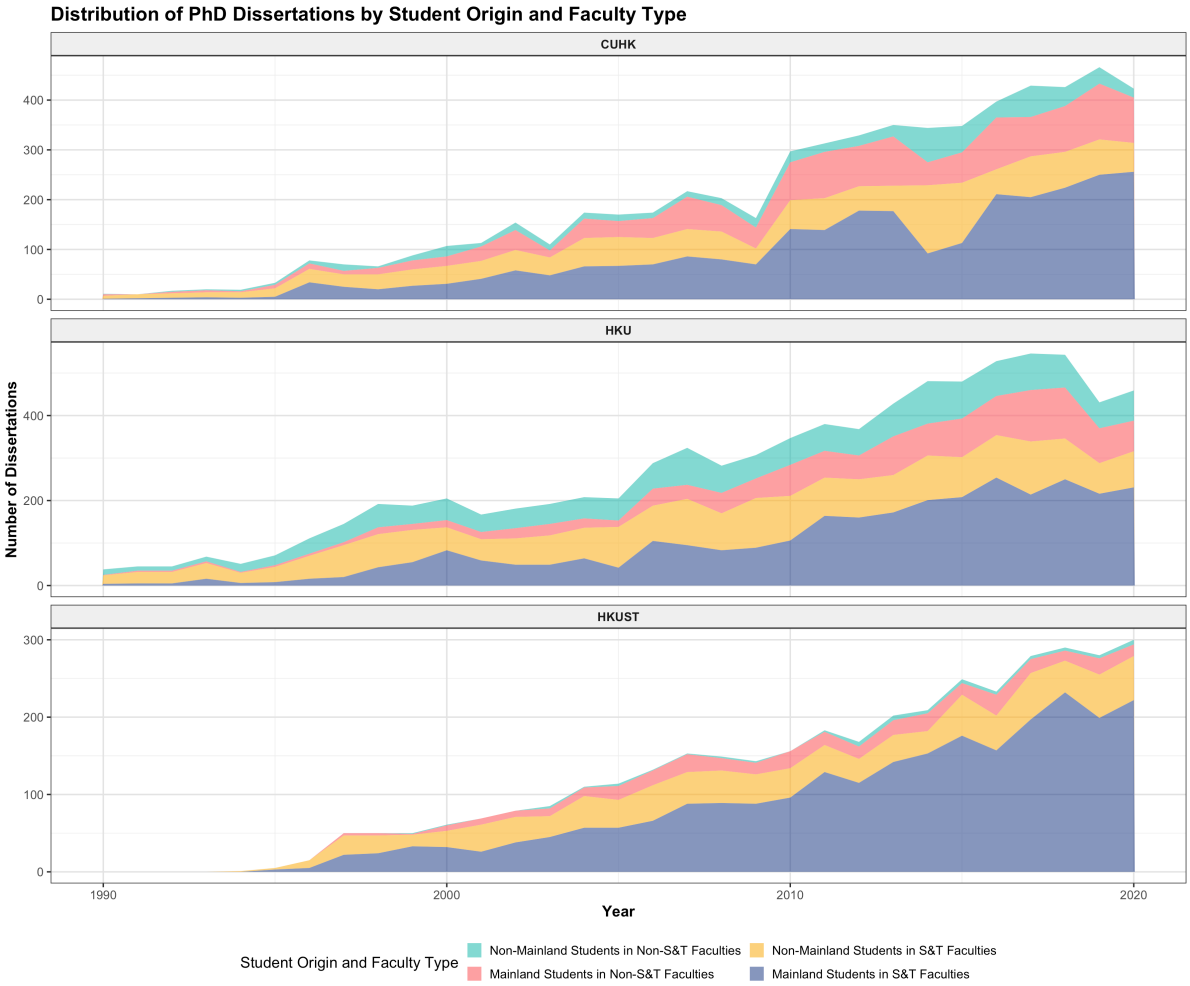


Figure 4: Distribution of PhD Dissertation by Student Origin and Faculty Type

The growth in Science and Technology PhD programs across Hong Kong's universities reveals an expansion in research capacity. At the established institutions, HKU and CUHK, S&T doctoral production accelerated markedly through the 1990s. HKU's annual S&T PhD graduates increased more than fivefold, from 24 in 1990 to 137 in 2000, while CUHK experienced nearly tenfold growth, from 7 to 67 graduates. HKUST, though newly established, demonstrated similar momentum, expanding from its inaugural PhD graduate in 1994 to 53 S&T PhDs by 2000.

The early 2000s brought varying growth patterns. HKU experienced a temporary plateau in graduation rates during 2000-2005, but subsequently recovered, reaching 211 S&T PhDs by 2010. In contrast, CUHK and HKUST maintained consistent growth trajectories. CUHK's graduation numbers rose to 125 PhDs in 2005 and 199 by 2010, while HKUST increased from 93 to 134 graduates during the same period.

Mainland Chinese students were central to this expansion. Their representation in S&T programs grew modestly in the early 1990s, with mainland students comprising 13% of HKU's and 36% of CUHK's S&T PhD graduates in 1990. The most significant shift occurred between 1995 and 2000, when mainland student representation approached parity at HKU (49%) and CUHK (47%). HKUST established a strong mainland presence from its inception, with mainland students representing 64% of its PhD graduates by 2000.

Considering the typical 3-4 year duration of Hong Kong PhD programs, these trends indicate that mainland students began entering S&T departments in the late 1980s. Their presence grew substantially through subsequent decades, reaching significant majorities by 2020: 66% at HKU, 82% at CUHK, and 79% at HKUST.

This timing is not coincidental, physicist Chien Chih-Yung noted in his memior that when he became the founding Dean of Science at HKUST in 1991, a survey of HKU and CUHK showed that only 5% of students were enrolled

in research PhD programs (Chien 2024, 5). By setting an ambitious target of 30% for research enrollment, HKUST signaled a transformative shift in Hong Kong's academic landscape. This shift paralleled with an interest to recruit mainland graduate students starting in mid 1980s by HKU and CUHK with and expanded to substantial intake beginning in 1995[16], creating a collaborative exchange that went beyond a simple transfer of knowledge. The development was seen as mutually beneficial as the local interest in S&T was limited and a vast untapped potential at the mainland side (Chan 2023, 80; Huang 2000; Dai 1997).

This collaboration yielded fruitful results, reflected not only in the increasing numbers of research-trained PhD graduates but also in the rise of publications in S&T fields. Here, we compare the three flagship universities from Hong Kong with top mainland institutions and graph their overall publication output in S&T fields from 1980 to 2020.

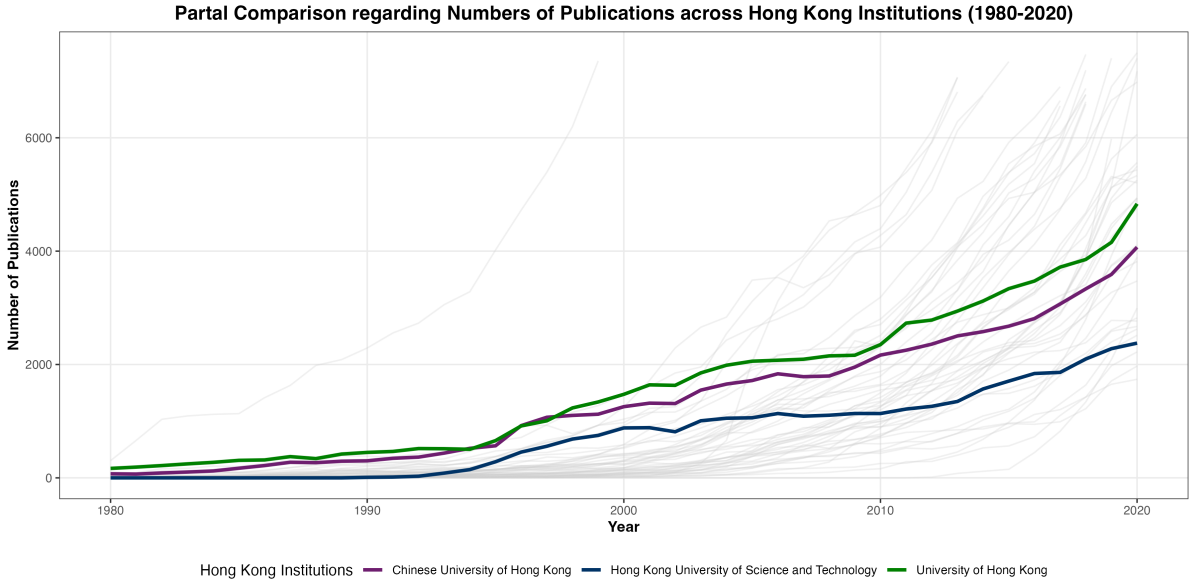


Figure 5: WoS publication trends of three Hong Kong universities and mainland institutions, 1980-2020)

From the 1980s to 1998 (the year after Hong Kong's handover to the PRC), the two leading universities, HKU and CUHK, outpaced all mainland universities in S&T research output, with the only exception being CAS[17]. HKUST, though new, quickly demonstrated its S&T prowess. Its growth throughout the 1990s surpassed that of most mainland

[16]The formal scholarly exchanges between Hong Kong and mainland universities commenced after 1978 (The Chinese University of Hong Kong 1982, 48). Private philanthropic initiatives played an important role. In 1981, entrepreneur Pao Yue-Kong provided HKD150,000 to Rayson Huang for mainland university visits (Huang 2000, 119; Liu 1993, 257). Prior to that, Huang also secured resources from the Croucher Foundation in 1979 to expand academic exchanges between Hong Kong and mainland institutions (Huang 2000, 127). In a parallel initiative, from 1982 onward, philanthropist Li Ka-Shing's contributions enabled scholars from Shantou to visit the University of Hong Kong (Liu 1993, 258). While the precise timing of mainland PhD students' first arrival at Hong Kong universities remains unrecorded, these privately funded scholarly visits established the groundwork for sustained academic exchange.

With regard to formal study, according to personal accounts, privately financed study was possible but remained very difficult in the mid-1980s (see Wang Yanxiang's essay in (Liu 1993), who enrolled at HKU in 1985). Mainland students often began enrolling in MPhil programs in S&T fields (see Cheng Jing's essay in (Liu 1993), who went to HKU in 1984; also see the account from Zhang Youqi, who was instrumental to bring mainland students to HKU (Chan 2023). This should not come as a surprise, as China only signed an agreement with the United States in 1979 to formalize student exchanges. Self-financed study abroad was officially announced in 1981 but remained under heavy administrative control. It wasn't until December 1984 that the Chinese government canceled its state review of individual applications to study abroad. The PhD dissertation data we compiled shows that the earliest record of a mainland student receiving a S&T PhD degree in the 1980s was Wu Zhenping in chemistry at HKU in 1987, followed by Ni Zhijie, also in chemistry at CUHK, in 1989. There were fewer than 10 PhD dissertations in S&T fields from mainland students across the two universities combined during the 1980s. In the case of HKUST, it welcomed its first batch of mainland graduate students soon after the establishment of the university in 1990–91 academic year (Ch'i 2014, 226).

[17]CAS played a unique role in the entire Chinese S&T landscape. Its output accounted for more than 30% of the mainland's total output in the 1980s, making it the undisputed "locomotive" of China's S&T enterprise during that period (Fan 2019). Also see next section.

The financial capabilities of Hong Kong universities significantly surpassed those of their mainland counterparts during this period. HKUST Vice President and biologist Kung Shain-Dow observed in 1996 that his institution's annual budget surpassed the

universities. The takeoff of mainland universities largely occurred between 2000 and 2010. From a publication perspective, Hong Kong's universities maintained their lead over most mainland universities from 1980 to 2015. No wonder they became a primary source of learning for mainland students.

The advanced development of Hong Kong's S&T sector—evidenced by rising publication counts and increasing numbers of PhD graduates—offered Chinese higher education institutions an attractive, accessible model with robust academic capacity and geographical proximity. With lower physical and cultural barriers compared to other international collaborators, Hong Kong provided mainland China with a readily accessible example of S&T excellence. Before mainland universities fully developed their own research capacities, Hong Kong's flourishing S&T sector served as a valuable learning resource for late-developing regions like China. Hong Kong's strong S&T performance, measured by publication volume and impact, attracted a growing number of mainland students and collaborators, establishing a virtuous cycle of academic exchange and collaboration.

6.3 Geographical characteristic

The scientific collaboration between Hong Kong and mainland China demonstrates distinct geographical patterns. Analysis of spatial distribution reveals two significant trends in Figure 6. Most notably, Beijing maintains a central position in these collaborations. Despite its geographical distance from Hong Kong, Beijing-based institutions produced more collaborative research with Hong Kong universities than any other mainland region. Data from CUHK, HKU, and HKUST indicate that Beijing contributed 13,416 co-authored papers from 1980 to 2023, exceeding Guangdong Province's 12,039 papers in second place, despite Guangdong's geographical proximity to Hong Kong.

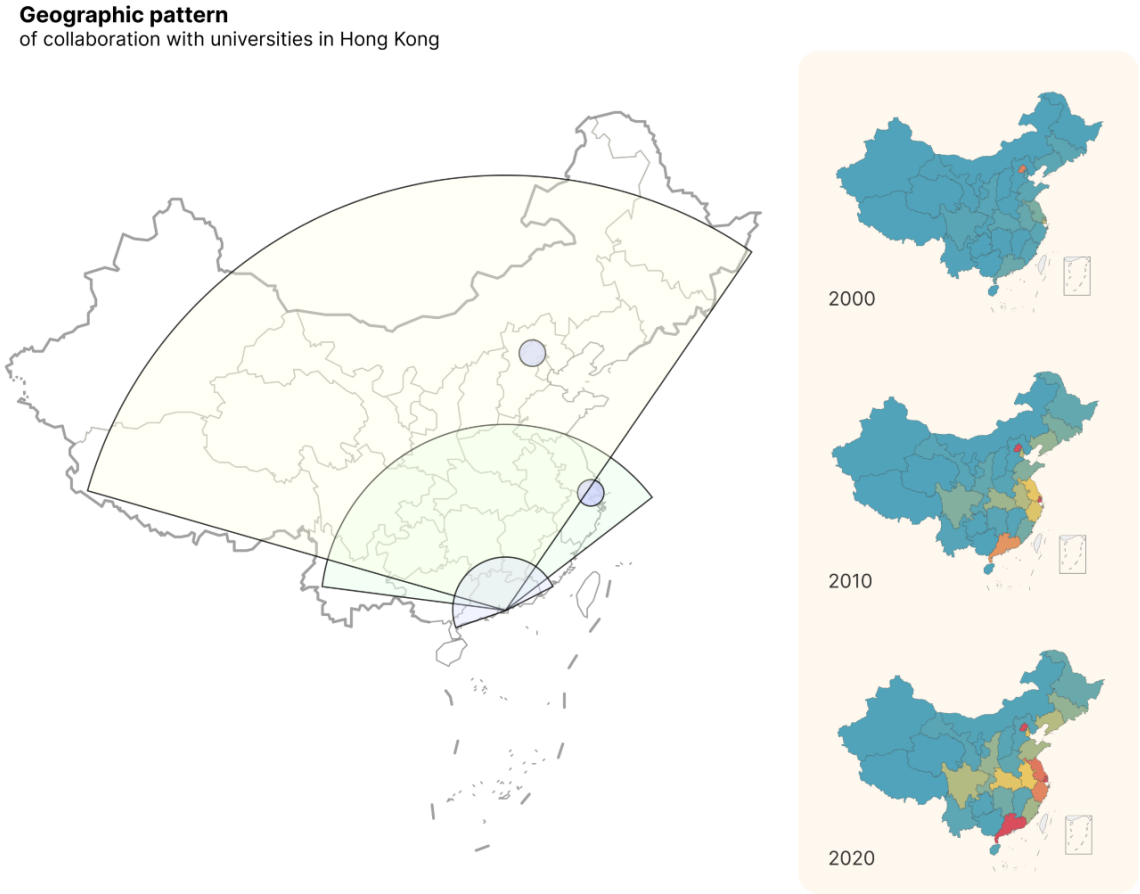


Figure 6: Partial comparison regarding numbers of publication across institutions (1990-2015)

combined resources allocated to three of mainland China's premier institutions: Peking University, Tsinghua University, and Fudan University (Kung 2000, 182). This substantial funding differential illustrates Hong Kong's capacity to provide advanced research support during mainland China's scientific development phase.

Beijing's primacy in China's scientific landscape stems from its status as home to the Chinese Academy of Sciences (CAS) and numerous prestigious universities. CAS has maintained its position as the nation's leading research institution since its founding. In the early 1980s, CAS generated more than one-third of China's total scientific output. While its proportional contribution decreased to approximately 10% of national scientific publications by 2018, this reduction reflects the expansion of China's research sector rather than diminished institutional influence.

Beyond its research output, CAS played a central role in developing China's scientific partnerships. The institution's leadership actively identified and established international and regional collaborations, providing strategic direction for China's scientific engagement globally. CAS consistently served as the initial point of contact for international scientists visiting China, fostering numerous long-term research partnerships.[18] The institution also pursued international connections proactively, establishing relationships that influenced the development of China's scientific networks.

A significant milestone in Hong Kong-mainland scientific collaboration occurred in 1991, during the early development of Hong Kong's research university system. CAS President Zhou Guangzhao's visits to CUHK and HKUST marked important steps in cross-border academic relations. CUHK awarded Zhou an honorary doctorate, while HKUST appointed him as the sole mainland representative on its Presidential Advisory Board. Zhou maintained active engagement, returning in September 1992 and participating in subsequent Advisory Board meetings in November 1992 and March 1994.[19]

Zhou's subsequent visit to HKU in 1995, where he received another honorary degree, reinforced CAS's commitment to academic partnerships. These visits extended beyond ceremonial significance, serving as strategic initiatives that prompted other CAS officials to establish research partnerships and joint laboratories with Hong Kong institutions.

The mid-1990s saw increased participation from CAS leadership. Vice Presidents Xu Zhihong, Teng Teng, Li Zhensheng, and Lu Yongxiang made multiple visits between 1993 and 1995. Notable among these visits was Lu Yongxiang's initial engagement while at Zhejiang University in May 1994, followed by his return visits as CAS Vice President. Additional high-level visits included Yan Yixun and Chen Yiyu in 1995, along with specialized research leaders such as Yan Dongsheng from the Shanghai Institute of Ceramics and Hong Zhongxiang from the Institute of Atmospheric Physics.

These visits culminated in the establishment of several joint research laboratories. The first, focusing on marine research, opened in Sanya, Hainan, in August 1994.[20] Subsequent laboratories were established in Life Science and Biotechnology (1996),[21] Microelectronics (January 1997), and Nanotechnology (November 1997). This progression demonstrates how institutional visits translated into concrete research collaborations across multiple scientific domains.

CAS has established extensive research partnerships across Hong Kong's higher education sector. By 2021, six Hong Kong universities maintained 22 joint laboratories with CAS, demonstrating the breadth of scientific collaboration. The geographical distribution of these laboratories reflects broader collaboration patterns: 11 laboratories operate in Guangdong Province, six in Beijing, two in Liaoning Province, and one each in Shanghai, Shaanxi Province, and Yunnan Province (Office of Hong Kong, Macao, and Taiwan Affairs, Chinese Academy of Sciences and Evaluation Research Center, Chinese Academy of Sciences 2022, 2).

The collaboration between CAS and Hong Kong universities exemplifies broader patterns in cross-border scientific partnerships. When examining these patterns excluding Beijing and Shanghai, geographical proximity emerges as a significant factor. Guangdong Province stands out as Hong Kong's primary regional research partner, with collabo-

[18]Examination of memoirs and biographies reveals that between 1970 and 1990, overseas Chinese scientists making their initial visits to the People's Republic of China consistently included the Chinese Academy of Sciences in their itineraries.

[19]Zhou presented a notable address at CUHK on October 24, 1991 (Zhou 1991). He also attended HKUST's opening ceremony in the same month. The Presidential Advisory Board at HKUST included distinguished international members: Morris Chang (Taiwan), Paolo Maria Fasella (Europe), Lord Flowers (UK), Donald Kennedy (USA), Fumio Hashimoto (Japan), Allen Lee (HK), Yuan T. Lee (USA), K.T. Li (Taiwan), Walter E. Massey (USA), Helmut Sohmen (HK), Chang-Lin Tien (USA), Lord Tombs of Brailes (UK), Zhou Guangzhao (PRC).

The data on CAS visits to HKUST derives from the university's Annual Reports published between academic years 1991-92 and 1997-98. The systematic documentation of these visits concluded in 1998.

[20]The laboratory received significant support from Hainan Province, a key partner in HKUST's mainland initiatives. The relationship between HKUST and Hainan extended beyond scientific collaboration, as demonstrated by the university's role in providing advanced training programs for Hainan government officials since 1992 (Ch'i 2014, 229). The laboratory project proved successful, with the CAS participating through its South China Sea Institute of Oceanology (SCSIO). CAS later established the Institute of Deep-sea Science and Engineering (IDSSE) in 2011. These three institutions, HKUST, SCSIO, and IDSSE, subsequently elevated the facility to the Sanya Comprehensive Marine Science Laboratory (Office of Hong Kong, Macao, and Taiwan Affairs, Chinese Academy of Sciences and Evaluation Research Center, Chinese Academy of Sciences 2022).

[21]This lab was not recorded in HKUST's annual report but was reported by Kung (Kung 2004, 62) and Zhang (1997).

ration intensity diminishing as geographical distance increases, as shown in Figure 6. This pattern was particularly pronounced between 1980 and 2000, when communication and travel limitations made partnerships with nearby regions more practical.

The development of academic partnerships between Hong Kong and Guangdong reflects the province's distinctive educational landscape. Despite its economic strength and substantial Hong Kong investment, Guangdong historically maintained limited research university infrastructure. Its primary scientific institutions comprised Sun Yat-Sen University, South China Normal University (with its teaching focus), Shenzhen University (founded 1983), and the newer South China University of Technology (founded 2011). This restricted academic capacity prompted Guangdong to build robust partnerships with Hong Kong universities, as evidenced by joint laboratory initiatives.

Figure 1 illustrates collaboration patterns across Guangdong's universities, revealing pronounced differences in partnership intensity. The data demonstrates exceptional levels of scientific and technological cooperation between Guangdong and Hong Kong institutions. Guangdong strategically deployed its economic resources to foster these academic relationships. In the 1990s, the province partnered with HKUST to establish industrial parks centered on collaborative research. This partnership model expanded when Guangdong invited Hong Kong universities to create satellite campuses, resulting in new CUHK (2014) and HKUST (2022) branches. These initiatives strengthened regional scientific networks while offering Guangdong access to Hong Kong's academic expertise, yielding advantages for both regions.

The pattern of scientific collaboration between Hong Kong and mainland China thus emerges from a synthesis of strategic institutional decisions and practical considerations, reflecting historical circumstances, geographical proximity, and technological capabilities.

7 Discussion

Universities in Hong Kong have left a rich historical legacy, documented through numerous studies of individual institutions and scientists. However, until now, there hasn't been a systematic examination of Hong Kong as a region and its relationship with mainland China in scientific development. Over the past four decades, we've seen a steady increase in both institutional and personal accounts. With the help of new data sources, this study provides the first comprehensive analysis of Hong Kong's higher education sector's regional impact on mainland China's S&T development.

A key question emerges: Was this development the result of a coordinated campaign? Current evidence suggests otherwise. The pattern appears more organic, driven by multiple factors and actors rather than centralized planning.

On the mainland side, the hunger for learning was intense after the Cultural Revolution, manifesting at both national and personal levels, albeit for different reasons. The pursuit of overseas education, whether for national advancement or personal growth, became a dominant aspiration among officials and young students alike. Students seized every opportunity for better education and career prospects, recognizing the stark disparities between mainland conditions and those of developed societies. Mainland policymakers and university leaders discovered Hong Kong's importance through direct visits, much as they had with Japan, Europe, and the United States from 1978 onward.

What distinguished Hong Kong was its unique position as a Chinese society with a high concentration of accomplished Chinese scientists. No other place globally offered such a confluence of Chinese-speaking scientific talent. The environment became particularly welcoming after the 1982 Sino-British Joint Declaration clarified Hong Kong's future. The city's advantages were obvious: no time-zone difference, no long-haul flights required, and no foreign language barriers (particularly at the CUHK and HKUST).

On Hong Kong's side, there is no evidence that universities coordinated their approach to mainland students before 2003, when China's Ministry of Education formally allowed Hong Kong universities to recruit mainland undergraduates. Instead, each institution developed its own outreach strategies. The biographical data reveals a consistent pattern: successive higher education leaders viewed science and technology as means to accelerate China's modernization, even when they didn't necessarily endorse the mainland regime. This perspective reflects a deeper historical current dating back to the 19th century, when China's technological backwardness first exposed its vulnerability to Western powers, and later reinforced by the devastating defeats in war after wars. For generations of Chinese intellectuals, these historical wounds underscored how scientific and technological weakness had left China unable to defend itself or realize its potential as a civilization. They saw science not merely as a path to modernization, but as essential to national survival and renewal.

For many overseas Chinese scientists, Hong Kong served as a waypoint home. The city offered a unique space where they could contribute to China's development while maintaining their academic freedom and international connections. This role challenges the conventional critique that Hong Kong neglected S&T in favor of commerce,

real estate, and finance. The data reveals a more nuanced picture, showing substantial investment and achievement in scientific research and education.

In many ways, this story represents an interlude in China's longer reemergence as a major power through S&T advancement. Hong Kong's universities, while maintaining their independence, provided crucial support during a critical period of mainland China's development. They offered not just scientific expertise and research facilities, but also a model of how Chinese institutions could engage with global science while maintaining their cultural identity.

Ma Lin, then CUHK president, foresaw this emerging partnership in 1984:

> I have not for a moment lost sight of the fact that Hong Kong is but a minute speck of sand in the ocean that is China. On the other hand, we must not forget, either, that throughout history a new idea or a wind of change often starts in a small, imperceptible way, then it gradually builds up its strength and eventually turns into a prevalent, irresistible movement which surprises us with its magnitude. Now that major changes in pursuit of modernization have already been set in motion in China, should not the people of Hong Kong, who have such a nose for following the wind ahead of the entire world, also rise to the occasion and contribute their share of zeal and spark? (Ma 1984)

Though small in size, Hong Kong cast a long shadow in China's scientific awakening, serving as both bridge and catalyst in the nation's journey toward technological ascendance.

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Appendix

A: Description of Bibliometric Data

Our bibliometric analysis draws from the Web of Science (WoS) Science Citation Index Expanded (SCI-E) database, examining publications from January 1980 through December 2023. We focused on three types of publications: *articles* (including journal articles, conference proceedings, and early-access articles), *reviews*, and *letters*.

The data collection and processing workflow, illustrated in Figure 7, began with automated queries to the WoS database. These queries extracted complete article metadata, with particular attention to author affiliations. Following the initial extraction, we converted the raw metadata into structured DataFrames, concatenating multiple export files and standardizing address fields to ensure consistency in our analysis.

A critical component of our methodology was the classification of author locations, detailed in Figure 8. We developed an algorithm to systematically categorize author affiliations into three groups based on their addresses: Hong Kong, mainland China, and international. For authors with multiple affiliations, we considered only the first listed address to ensure consistent classification. This process allowed us to determine the total number of authors by institution or by country / region.

The scope of our dataset is depicted in Figure 9. From the total Web of Science (WoS) universe of 43,649,369 publications, we identified 6,324,065 publications with at least one mainland China-based author and 310,463 publications with at least one Hong Kong-based author. We designate these two groups as B and H respectively. The Hong Kong group (H) constitutes our primary analytical dataset.

For Hong Kong papers, we divide them into two subgroups: h_1, papers whose authors are all from Hong Kong, and h_2, papers with at least one outside collaborating author (i.e., papers with international collaboration).

The h_1 group can be further decomposed into h_{1a} (single-author papers) and h_{1b} (multi-author papers where all authors are from Hong Kong).

The h_2 group can be further decomposed into h_{2a} (papers where the outside collaborator(s) are exclusively from mainland China), h_{2b} (papers with at least one mainland China-based author AND at least one author from another country/region), and h_{2o} (papers where the outside collaborator(s) are from countries other than mainland China). Therefore, we have $h_1 = h_{1a} + h_{1b}$, $h_2 = h_{2a} + h_{2b} + h_{2o}$, and $H = h_1 + h_2$.

We define similar groups for mainland China papers. We have $b_1 = b_{1a} + b_{1b}$, $b_2 = b_{2a} + b_{2b} + b_{2o}$, and $B = b_1 + b_2$. The key difference is that for mainland China, we highlight its collaborative relationship with the U.S., so in b_{2a}, b_{2b}, and b_{2o}, the outside collaborator(s) include U.S.-based authors.

It's important to note that there is no singular method to assign a geographic location to a scientific paper, as scientific collaboration transcends national boundaries. In fact, collaboration has become the norm in modern science. While researchers in different fields adopt various approaches to signal their respective contributions, there is no standardized way to document this division of labor, and WoS data does not provide such metrics. Therefore, our approach follows the current field standard by adopting a plural conception of authorship. Most papers, except those with single authors, can be simultaneously classified as belonging to multiple countries. Consequently, "Chinese papers" and "American papers" are not mutually exclusive sets.

To understand institutional collaborative patterns (Figure 1), we then developed a Collaboration Score (CS) to quantify the strength of international research partnerships. For an institution I collaborating with country/region C, the score is calculated as:

$$CS_{I,C} = \frac{n_C}{N_I}$$

where n_C represents the number of authors from country C in collaborative publications, and N_I denotes the total number of non-domestic authors across all institution I's collaborative papers. For instance, a CS of 0.1 indicates that authors from country C constitute 10% of all international authorship in institution I's collaborative publications. This metric provides an intuitive measure for analyzing large-scale international research collaboration patterns and identifying key partnership strengths across geographic regions.

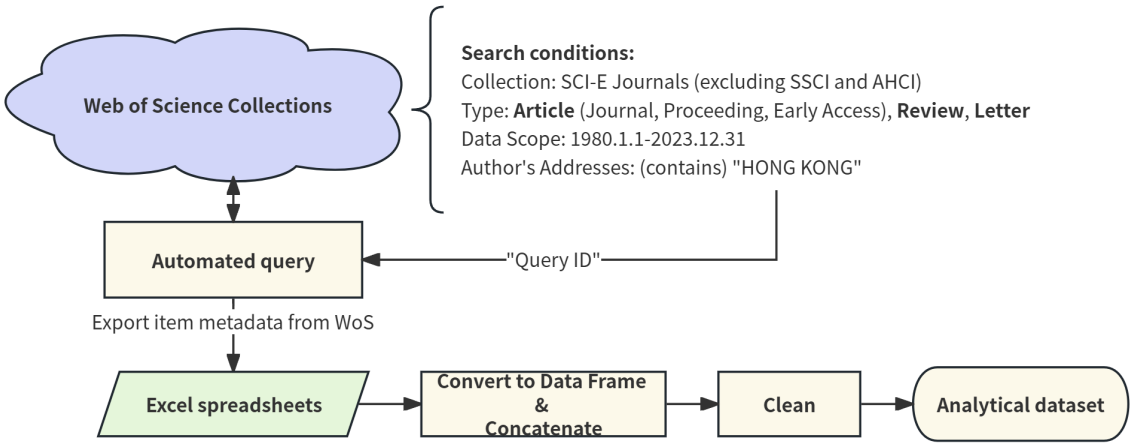

Figure 7: Steps in Collecting Web of Science Data

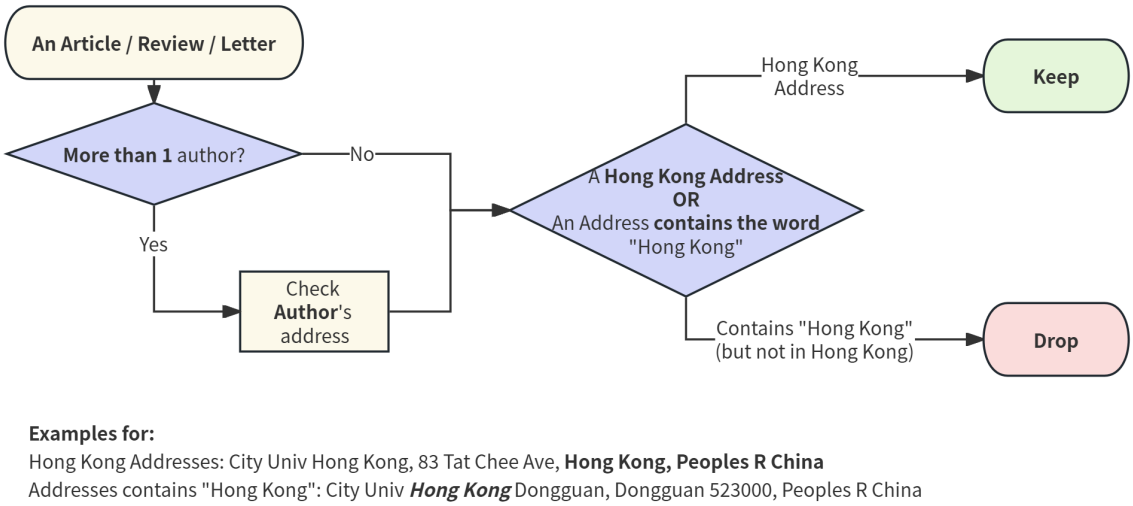

Figure 8: Determining Location of WoS Entries

B: Chinese Universities Included for Analysis

Our analysis includes three major Hong Kong universities: the University of Hong Kong (HKU), the Chinese University of Hong Kong (CUHK), and the Hong Kong University of Science and Technology (HKUST), as well as selected universities from mainland China.

For mainland universities, we used the *Nature Index* 2024 China ranking as our primary selection criterion for top S&T performers. From the top 50 institutions in this ranking, we excluded the three Hong Kong universities mentioned above, plus the City University of Hong Kong (37) and the People's Liberation Army (42), resulting in 45 mainland Chinese universities.

Our analysis expanded the institutional sample to account for the geographic significance of Guangdong Province's proximity to Hong Kong. The final sample comprises 62 universities, constructed from two groups: the top 45 S&T universities in China (of which five are located in Guangdong) and an additional 17 Guangdong institutions. This comprehensive approach includes all 22 universities in Guangdong Province as of 2023, enabling thorough analysis of regional collaboration patterns.

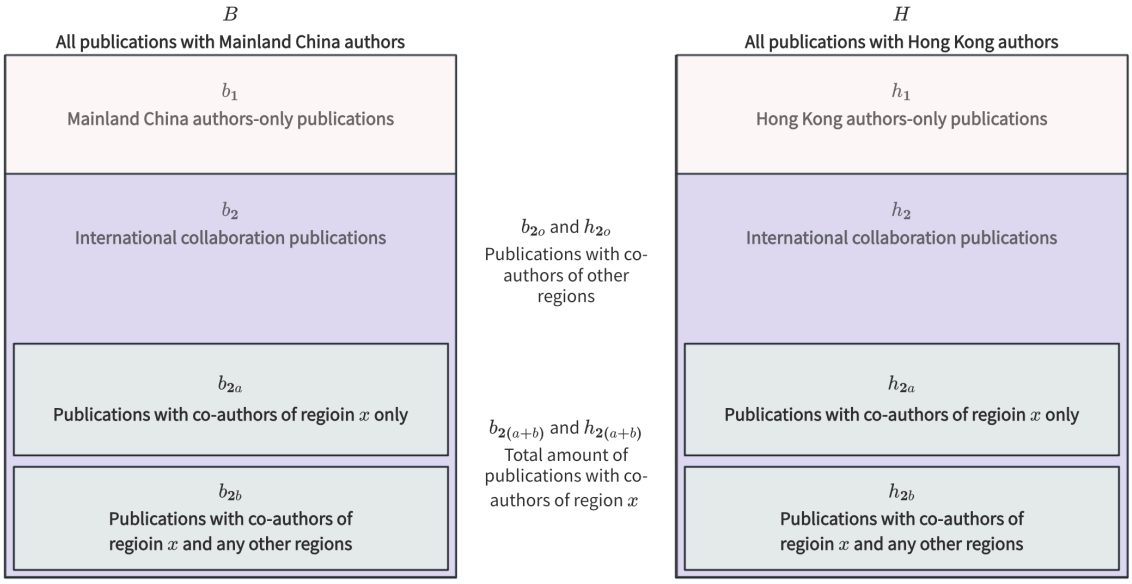

Figure 9: Universe of Publications and Collaborations

Table A1: Publication Data Breakdown for mainland China and Hong Kong SAR

Year	Mainland China									Hong Kong								
	B	b_1			b_2 (e.g. x=USA)					H	h_1			h_2 (e.g. x=Mainland)				
		total	b_{1a}	b_{1b}	total	b_{2a}	b_{2b}	$b_{2a}+b_{2b}$	b_{2o}		total	h_{1a}	h_{1b}	total	h_{2a}	h_{2b}	$h_{2a}+h_{2b}$	h_{2o}
2023	696291	567261			129030	25222	12708	37930	91100	24535	3240	190	3050	21295	9535	6976	16511	4784
2022	716299	574634			141665	31147	13911	45058	96607	24255	3641	225	3416	20614	9230	6384	15614	5000
2021	617277	472872			144405	35957	14995	50952	93453	21644	3605	219	3386	18039	7693	5572	13265	4774
2020	532929	397270			135659	38455	14533	52988	82671	18696	3116	203	2913	15580	6649	4699	11348	4232
2019	474269	352374			121895	38637	13524	52161	69734	16466	2781	188	2593	13685	5837	4105	9942	3743
2018	386428	284918			101510	34387	11189	45576	55934	15196	2932	231	2701	12264	5179	3588	8767	3497
2017	335609	248793			86816	31114	9605	40719	46097	13633	2846	199	2647	10787	4494	2978	7472	3315
2016	300591	224497			76094	27803	8603	36406	39688	12639	2722	226	2496	9917	4330	2648	6978	2939
2015	272812	206776			66036	24606	7215	31821	34215	11981	3024	249	2775	8957	4405	1928	6333	2624
2014	242584	185428			57156	20939	6063	27002	30154	11434	3157	229	2928	8277	4110	1638	5748	2529
2013	209201	160073			49128	18157	5167	23324	25804	10901	3335	233	3102	7566	3689	1345	5034	2532
2012	175985	134879			41106	14820	4230	19050	22056	10136	3286	211	3075	6850	3363	1161	4524	2326
2011	151226	115661			35565	12589	3428	16017	19548	9720	3367	253	3114	6353	3131	1051	4182	2171
2010	128819	98650			30169	10448	2819	13267	16902	8938	3261	267	2994	5677	2705	869	3574	2103
2009	116496	90796			25700	8557	2259	10816	14884	8526	3168	250	2918	5358	2612	754	3366	1992
2008	100113	78392			21721	6921	1880	8801	12920	8059	3252	252	3000	4807	2267	670	2937	1870
2007	86910	68920			17990	5561	1533	7094	10896	7807	3148	255	2893	4659	2270	571	2841	1818
2006	77518	61417			16101	4804	1327	6131	9970	7980	3509	228	3281	4471	2201	513	2714	1757
2005	64160	50766			13394	3902	1163	5065	8329	7591	3440	245	3195	4151	2069	438	2507	1644
2004	51875	40461			11414	3127	952	4079	7335	7299	3454	256	3198	3845	1852	393	2245	1600
2003	40793	31487			9306	2581	748	3329	5977	6939	3391	326	3065	3548	1638	360	1998	1550
2002	33393	25548			7845	2052	673	2725	5120	5940	2862	261	2601	3078	1445	282	1727	1351
2001	29374	22668			6706	1736	560	2296	4410	5861	3041	309	2732	2820	1267	256	1523	1297
2000	24776	19072			5704	1422	470	1892	3812	5329	2637	289	2348	2692	1221	254	1475	1217
1999	21737	16882			4855	1168	382	1550	3305	4604	1631	203	1428	2973	1753	343	2096	877
1998	18345	14023			4322	1130	333	1463	2859	4158	1988	282	1706	2170	1058	210	1268	902
1997	16168	12641			3527	837	308	1145	2382	3630	2218	364	1854	1412	534	83	617	795
1996	14450	11418			3032	798	278	1076	1956	2984	1898	377	1521	1086	328	38	366	720
1995	11990	9476			2514	696	221	917	1597	1988	1264	293	971	724	185	43	228	496
1994	9618	7439			2179	577	180	757	1422	1563	1016	207	809	547	114	13	127	420
1993	8887	6793			2094	549	204	753	1341	1358	914	181	733	444	83	13	96	348
1992	8346	6411			1935	623	178	801	1134	1151	831	157	674	320	69	4	73	247
1991	7600	5819			1781	604	133	737	1044	1001	773	154	619	228	44	7	51	177
1990	7195	5706			1489	540	127	667	822	971	774	163	611	197	34	4	38	159
1989	6539	5302			1237	485	115	600	637	920	765	144	621	155	24	1	25	130
1988	5881	4783			1098	443	86	529	569	749	610	152	458	139	21	6	27	112
1987	4874	4096			778	364	44	408	370	770	618	141	477	152	14	1	15	137
1986	4015	3266			749	340	42	382	367	639	512	109	403	127	15	6	21	106
1985	3170	2490			680	330	38	368	312	562	441	121	320	121	10	2	12	109
1984	2987	2353			634	317	36	353	281	512	401	135	266	111	7	0	7	104
1983	2753	2245			508	261	30	291	217	394	310	87	223	84	5	1	6	78
1982	2375	2060			315	155	21	176	139	382	311	87	224	71	6	0	6	65
1981	1491	1319			172	81	11	92	80	319	253	87	166	66	1	0	1	65
1980	767	698			69	32	8	40	29	303	245	81	164	58	1	0	1	57

Table B1: Table: Chinese University Samples

Nature China Ranking	Institution	Sources
1	Chinese Academy of Sciences	Global Top 50
2	University of Chinese Academy of Sciences (UCAS)	Global Top 50
3	University of Science & Technology of China (USTC)	Global Top 50
4	Peking University	Global Top 50
5	Nanjing University	Global Top 50
6	Zhejiang University	Global Top 50
7	Tsinghua University	Global Top 50
8	Sun Yat Sen University	Global Top 50 in Guangdong
9	Shanghai Jiao Tong University	Global Top 50
10	Fudan University	Global Top 50
11	Sichuan University	Global Top 50
12	Nankai University	Global Top 50
13	Wuhan University	Global Top 50
14	Huazhong University of Science & Technology	Global Top 50
15	Shandong University	Global Top 50
16	Xiamen University	Global Top 50
17	Soochow University - China	Global Top 50
18	Southern University of Science & Technology	Global Top 50 in Guangdong
19	Jilin University	Global Top 50
20	Central South University	Global Top 50
21	Tianjin University	Global Top 50
22	Hunan University	Global Top 50
23	Xi'an Jiaotong University	China Top 50
24	Harbin Institute of Technology	China Top 50
25	South China University of Technology	China Top 50 in Guangdong
26	Tongji University	China Top 50
27	Lanzhou University	China Top 50
28	Zhengzhou University	China Top 50
29	Beijing Normal University	China Top 50
30	Beijing Institute of Technology	China Top 50
31	Chongqing University	China Top 50
33	Shenzhen University	China Top 50 in Guangdong
34	Southeast University	China Top 50
35	Dalian University of Technology	China Top 50
36	Beijing University of Chemical Technology	China Top 50
38	East China University of Science & Technology	China Top 50
39	East China Normal University	China Top 50
43	Beihang University	China Top 50
44	China University of Geosciences	China Top 50
45	Northwestern Polytechnical University	China Top 50
46	Jinan University	China Top 50 in Guangdong
47	Fuzhou University	China Top 50
48	Peking Union Medical College (CAMS)	China Top 50
49	Nanjing Tech University	China Top 50
50	Shaanxi Normal University	China Top 50
52	Southern Medical University	China Top 100 in Guangdong
68	Guangdong University of Technology	China Top 100 in Guangdong
74	South China Normal University	China Top 100 in Guangdong
117	South China Agricultural University	in Guangdong
Not Ranked	Guangzhou University	in Guangdong
Not Ranked	Shantou University	in Guangdong
Not Ranked	Shenzhen Technology University	in Guangdong
Not Ranked	Guangzhou Medical University	in Guangdong
Not Ranked	Foshan University	in Guangdong
Not Ranked	Wuyi University	in Guangdong
Not Ranked	Huizhou University	in Guangdong

Nature China Ranking	Institution	Sources
Not Ranked	Guangzhou University of Chinese Medicine	in Guangdong
Not Ranked	Guangdong University of Petrochemical Technology	in Guangdong
Not Ranked	Zhongkai University of Agriculture & Engineering	in Guangdong
Not Ranked	Zhaoqing University	in Guangdong
Not Ranked	Shenzhen Institute of Information Technology	in Guangdong
Not Ranked	Shunde Polytechnic	in Guangdong

C: Science and Technology Research Areas

The 110 Web of Science research categories were consolidated into 27 meso-categories through ontological grouping and manual adjustment.

Initial clusters were formed using hierarchical domain ontologies:

- **Medical Subject Headings (MeSH)** tree structures guided biomedical category mergers (e.g., grouping Cardiology and Hematology under Clinical Medicine Specialties)
- **OECD Frascati Manual 2020** R&D classifications informed technology and physical science groupings (e.g., Energy Technologies cluster)
- **ESCO Taxonomy** (European Skills/Competences framework) validated engineering and materials science clusters

We have 4 large groups: 1) Life sciences and biomedicine (14 meso-category); 2) Physical Sciences (6 meso-category); 3) Technology (5 meso-category); 4) Unclassified topics (2 meso-category). This structure maintains the original category names and reduce the granularity by 77%. The meso-category preserves WoS's native superdomain structure (Life Sciences/Physical/Tech).

Table B2: Wos Research Topics - Meso Level Research Category Mapping

Original Category	Meso-Category
Agriculture	Agriculture & Food Systems
Allergy	Clinical Medicine Specialties
Anatomy & Morphology	Basic Biological Mechanisms
Anesthesiology	Clinical Support Sciences
Anthropology	Neuroscience & Behavior
Audiology & Speech-Language Pathology	Clinical Support Sciences
Behavioral Sciences	Neuroscience & Behavior
Biochemistry & Molecular Biology	Basic Biological Mechanisms
Biodiversity & Conservation	Evolution & Biodiversity
Biophysics	Basic Biological Mechanisms
Biotechnology & Applied Microbiology	Biomedical Technology
Cardiovascular System & Cardiology	Clinical Medicine Specialties
Cell Biology	Basic Biological Mechanisms
Critical Care Medicine	Clinical Support Sciences
Dentistry, Oral Surgery & Medicine	Clinical Support Sciences
Dermatology	Clinical Medicine Specialties
Developmental Biology	Aging & Development
Emergency Medicine	Clinical Support Sciences
Endocrinology & Metabolism	Clinical Medicine Specialties
Entomology	Evolution & Biodiversity
Environmental Sciences & Ecology	Environmental Health
Evolutionary Biology	Evolution & Biodiversity
Fisheries	Agriculture & Food Systems
Food Science & Technology	Agriculture & Food Systems
Forestry	Agriculture & Food Systems
Gastroenterology & Hepatology	Clinical Medicine Specialties
General & Internal Medicine	Clinical Medicine Specialties
Genetics & Heredity	Basic Biological Mechanisms
Geriatrics & Gerontology	Aging & Development
Health Care Sciences & Services	Public Health Systems
Hematology	Clinical Medicine Specialties
Immunology	Infection & Immunity
Infectious Diseases	Infection & Immunity
Integrative & Complementary Medicine	Alternative Medicine
Legal Medicine	Public Health Systems
Marine & Freshwater Biology	Environmental Health

Original Category	Meso-Category
Mathematical & Computational Biology	Mathematical Sciences
Medical Ethics	Public Health Systems
Medical Informatics	Biomedical Technology
Medical Laboratory Technology	Diagnostic Medicine
Microbiology	Infection & Immunity
Mycology	Evolution & Biodiversity
Neurosciences & Neurology	Neuroscience & Behavior
Nursing	Clinical Support Sciences
Nutrition & Dietetics	Agriculture & Food Systems
Obstetrics & Gynecology	Clinical Medicine Specialties
Oncology	Clinical Medicine Specialties
Ophthalmology	Surgical Disciplines
Orthopedics	Clinical Medicine Specialties
Otorhinolaryngology	Surgical Disciplines
Paleontology	Evolution & Biodiversity
Parasitology	Infection & Immunity
Pathology	Diagnostic Medicine
Pediatrics	Clinical Medicine Specialties
Pharmacology & Pharmacy	Clinical Support Sciences
Physiology	Basic Biological Mechanisms
Plant Sciences	Evolution & Biodiversity
Psychiatry	Neuroscience & Behavior
Public, Environmental & Occupational Health	Public Health Systems
Radiology, Nuclear Med & Medical Imaging	Biomedical Technology
Rehabilitation	Alternative Medicine
Reproductive Biology	Aging & Development
Research & Experimental Medicine	Clinical Support Sciences
Respiratory System	Clinical Medicine Specialties
Rheumatology	Clinical Medicine Specialties
Sport Sciences	Alternative Medicine
Substance Abuse	Public Health Systems
Surgery	Surgical Disciplines
Toxicology	Environmental Health
Transplantation	Surgical Disciplines
Tropical Medicine	Public Health Systems
Urology & Nephrology	Surgical Disciplines
Veterinary Sciences	Agriculture & Food Systems
Virology	Infection & Immunity
Zoology	Evolution & Biodiversity
Astronomy & Astrophysics	Astrophysics & Space Science
Chemistry	Fundamental Chemistry
Crystallography	Fundamental Chemistry
Electrochemistry	Fundamental Chemistry
Geochemistry & Geophysics	Geosciences
Geology	Geosciences
Mathematics	Mathematical Sciences
Meteorology & Atmospheric Sciences	Astrophysics & Space Science
Mineralogy	Geosciences
Mining & Mineral Processing	Energy Technologies
Oceanography	Astrophysics & Space Science
Optics	Applied Physics
Physical Geography	Earth Systems
Physics	Applied Physics
Polymer Science	Fundamental Chemistry
Thermodynamics	Applied Physics
Water Resources	Earth Systems
Acoustics	Analytical Instrumentation

Original Category	**Meso-Category**
Automation & Control Systems	Engineering Systems
Computer Science	Information Systems
Construction & Building Technology	Materials Science
Energy & Fuels	Energy Technologies
Engineering	Engineering Systems
Imaging Science & Photographic Technology	Analytical Instrumentation
Information Science & Library Science	Information Systems
Instruments & Instrumentation	Analytical Instrumentation
Materials Science	Materials Science
Mechanics	Engineering Systems
Metallurgy & Metallurgical Engineering	Materials Science
Microscopy	Analytical Instrumentation
Nuclear Science & Technology	Energy Technologies
Operations Research & Management Science	Mathematical Sciences
Remote Sensing	Information Systems
Robotics	Engineering Systems
Science & Technology Other Topics	Information Systems
Spectroscopy	Analytical Instrumentation
Telecommunications	Information Systems
Transportation	Engineering Systems
Life Sciences & Biomedicine - Other Topics	Other Life Sciences & Biomedicine Topics
Science & Technology - Other Topics	Other Science & Technology Topics

Compared to the commonly used OECD classification fields[22], which has 6 broad fields with 42 second-level subfields, these meso-level categories map nicely:

Table B3: Meso Level Research Category - OECD classification mapping

Meso-Category	**OECD Field of R&D Classification** (Second-Level)
Basic Biological Mechanisms	1.6 Biological sciences
Clinical Medicine Specialties	3.2 Clinical medicine
Public Health Systems	3.3 Health sciences
Environmental Health	2.7 Environmental engineering, 1.5 Earth and related environ sciences
Neuroscience & Behavior	5.1 Psychology and cognitive sciences
Biomedical Technology	2.6 Medical engineering
Evolution & Biodiversity	1.6 Biological sciences
Agriculture & Food Systems	4.1 Agriculture, forestry, and fisheries
Infection & Immunity	3.1 Basic medicine
Aging & Development	3.1 Basic medicine
Clinical Support Sciences	3.2 Clinical medicine
Diagnostic Medicine	3.2 Clinical medicine
Surgical Disciplines	3.2 Clinical medicine
Astrophysics & Space Science	1.3 Physical sciences
Geosciences	1.5 Earth and related environmental sciences
Fundamental Chemistry	1.4 Chemical sciences
Applied Physics	1.3 Physical sciences
Mathematical Sciences	1.1 Mathematics
Earth Systems	1.5 Earth and related environmental sciences
Engineering Systems	2.3 Mechanical engineering, 2.2 Electrical engineering
Materials Science	2.5 Materials engineering
Energy Technologies	2.8 Environmental biotechnology, 2.9 Industrial biotechnology
Information Systems	1.2 Computer and information sciences
Analytical Instrumentation	Not explicitly covered in OECD fields

[22] https://www.oecd.org/content/dam/oecd/en/publications/reports/2015/10/frascati-manual-2015_g1g57dcb/9789264239012-en.pdf

Meso-Category	**OECD Field of R&D Classification** (Second-Level)
Alternative Medicine	Not explicitly covered in OECD fields

The two exceptions are specialty areas that Chinese and Hong Kong institutions engaged in.

D: Number of PhD dissertations in S&T sector for HKU, CUHK and HKUST (1990-2020)

The dissertation data were collected from the library systems of three universities: the University of Hong Kong (HKU), the Chinese University of Hong Kong (CUHK), and the Hong Kong University of Science and Technology (HKUST). For each dissertation, we documented four key variables: author, year, title, and department/faculty affiliation. To align with our research focus on science and technology (S&T), we limited our dataset to dissertations from relevant fields.

The original data contained different departmental classification systems across the three universities: HKU used "discipline" labels, HKUST used "department" designations, and CUHK used "division" categories. To standardize these classifications, we mapped each departmental unit to its position within the broader organizational structure of its respective university. This mapping enabled us to aggregate the departmental-level information to faculty-level categories for consistent cross-institutional analysis.

For example, the Faculty of Engineering at CUHK would contain these divisions:

- Division of Automation and Computer-Aided Engineering
- Division of Biomedical Engineering
- Division of Computer Science and Engineering
- Division of Computer Science
- Division of Electronic Engineering
- Division of Electronics
- Division of Information Engineering
- Division of Materials Science and Engineering
- Division of Mechanical and Automation Engineering
- Division of Systems Engineering and Engineering Management

Faculty of Science would include:

- Division of Biochemistry
- Division of Biology
- Division of Chemistry
- Division of Earth and Atmospheric Sciences
- Division of Earth System and GeoInformation Science
- Division of Environmental Science
- Division of Life Sciences
- Division of Mathematics
- Division of Microbiology
- Division of Molecular Biotechnology
- Division of Physics
- Division of Statistics

Similary, Faculty of Medicine would include:

- Division of Anaesthesia and Intensive Care.
- Division of Anatomical and Cellular Pathology.
- Division of Anatomy.
- Division of Chemical Pathology.
- Division of Chinese Medicine.
- Division of Clinical and Pathological Sciences.
- Division of Diagnostic Radiology and Organ Imaging.
- Division of Imaging and Interventional Radiology.
- Division of Medical Sciences.
- Division of Medicine and Therapeutics.
- Division of Medicine.

- Division of Nursing.
- Division of Obstetrics and Gynaecology.
- Division of Ophthalmology and Visual Sciences.
- Division of Orthopaedics and Traumatology.
- Division of Pathological Sciences.
- Division of Pharmacology.
- Division of Pharmacy.
- Division of Physiology.
- Division of Public Health.
- Division of Surgery.
- Division of Surgical Sciences.

For HKU, Faculty of Engineering would include:

- Civil and Structural Engineering
- Civil Engineering
- Computer Science
- Computer Science and Information Systems
- Electrical and Electronic Engineering
- Electrical Engineering
- Industrial and Manufacturing Systems Engineering
- Industrial Engineering
- Mechanical Engineering

Facult of Science would include:

- Biochemistry
- Biological Sciences
- Botany
- Chemistry
- Earth Sciences
- Ecology and Biodiversity
- Mathematics
- Microbiology
- Molecular Biology
- Oral Biology
- Physics
- Statistics
- Statistics and Actuarial Science
- Zoology

Faculty of Medicine would include:

- Anaesthesiology
- Anatomy
- Biomedical Sciences
- Chinese Medicine
- Clinical Oncology
- Clinical Psychology
- Community Medicine
- Diagnostic Radiology
- Family Medicine and Primary Care
- Humanities and Medicine
- Medicine
- Nursing Studies
- Obstetrics and Gynaecology
- Ophthalmology
- Orthopaedic Surgery
- Orthopaedics and Traumatology
- Paediatrics

- Paediatrics and Adolescent Medicine
- Pathology
- Pharmacology
- Pharmacology and Pharmacy
- Physiology
- Psychiatry
- Public Health
- Radioisotope
- Radioisotope Techniques
- Surgery

For HKUST, Faculty of Engineering would include:

- Bioengineering
- Biomedical Engineering
- Chemical and Biological Engineering
- Chemical and Biomolecular Engineering
- Chemical Engineering
- Civil and Environmental Engineering
- Civil and Structural Engineering
- Civil Engineering
- Computer Science
- Computer Science and Engineering
- Electrical and Electronic Engineering
- Electronic and Computer Engineering
- Environmental Engineering
- Industrial Engineering and Decision Analytics
- Industrial Engineering and Engineering Management
- Industrial Engineering and Logistics Management
- Mechanical and Aerospace Engineering
- Mechanical Engineering

Faculty of Science would include:

- Atmospheric, Marine and Coastal Environment
- Biochemistry
- Biology
- Chemistry
- Environment
- Environment and Sustainability
- Environment and Sustainability Atmospheric Environmental Science
- Environment and Sustainability Environmental Science, Policy and Management
- Environmental Science
- Life Science
- Life Science – Biochemistry and Cell Biology
- Life Science – Marine Ecology and Biotechnology
- Mathematics
- Nano Science and Technology
- Ocean Science
- Physics
- Physics Nano Science and Technology

To focus on science and technology fields, we selected specific faculties from each university. From HKU and CUHK, we included the Faculty of Science, Faculty of Engineering, and Faculty of Medicine. From HKUST, we included the Faculty of Science and Faculty of Engineering. Our analysis excluded all faculties in social sciences, humanities, architecture, arts, business, economics, law, and education.

To estimate the proportion of mainland graduates, we analyzed students' names based on their romanization patterns. Hong Kong students typically use Cantonese romanization, mainland students use Pinyin romanization, and Taiwanese students generally use Wade-Giles or other romanization systems.

The analytical period spans from 1990 to 2020, accommodating several institutional constraints. First, the universities have different histories of PhD programs: HKU awarded its first PhD in 1981, while HKUST graduated its first PhD cohort in 1995. Second, library systems maintain different embargo periods, with CUHK's PhD database having complete records only until 2020. The exclusion of pre-1990 data has minimal impact on our analysis, as both HKU and CUHK produced fewer than ten PhD graduates annually in S&T fields during this period.

Table D1: PhD dissertations in S&T fields for HKU, CUHK and HKUST from 1990 to 2020.

Year	All PhD dissertation			S&T dissertation			Mainland S&T estimates		
	HKU	CUHK	HKUST	HKU	CUHK	HKUST	HKU	CUHK	HKUST
1990	38	11	NA	24	7	NA	4	1	NA
1991	45	10	NA	32	10	NA	5	2	NA
1992	45	17	NA	32	12	NA	5	3	NA
1993	68	20	NA	53	14	NA	16	4	NA
1994	51	19	1	30	14	1	6	3	0
1995	71	33	5	44	22	5	8	5	3
1996	111	78	15	70	61	15	16	34	5
1997	145	70	50	95	50	47	20	25	22
1998	192	66	50	121	50	47	43	20	24
1999	188	88	50	131	60	48	55	27	33
2000	205	107	61	137	67	53	83	31	32
2001	167	113	69	109	77	61	59	41	26
2002	181	154	79	111	99	71	49	58	38
2003	192	110	85	118	84	72	49	48	45
2004	208	174	110	136	123	98	64	66	57
2005	205	170	114	138	125	93	42	67	57
2006	288	174	132	188	123	112	105	70	66
2007	324	217	153	204	141	129	95	86	88
2008	282	203	149	170	136	131	83	80	89
2009	307	163	143	206	102	126	89	70	88
2010	347	297	156	211	199	134	106	141	96
2011	380	313	183	254	203	164	164	139	129
2012	368	329	168	250	227	146	160	178	115
2013	428	350	202	260	228	177	172	177	142
2014	481	344	209	306	229	182	201	92	153
2015	480	348	249	302	234	229	208	113	176
2016	528	397	233	354	261	202	254	211	157
2017	546	429	279	339	287	257	214	205	197
2018	543	426	290	346	296	273	250	224	232
2019	431	466	280	288	321	255	216	250	199
2020	459	423	300	316	314	279	231	256	222

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